

### COMBINED Series

1. Photometer is used in determining

- A. ☒ Intensity of light
- B. Temperature
- C. Pressure
- D. Mass

2. How many ionisable hydrogen ions can be found in acetic acid?

- A. 1 ☒
- B. 2
- C. 3
- D. 0

3. Who discovered the protoplasm in the cell?

- A. Schleiden
- B. Schwann
- C. ☒ Dujardin
- D. Hooke

4. Which of the following is not a use of electroscope?

- A. Detecting electric charge
- B. Measuring potential difference
- C. Detecting ionizing radiation
- D. ☒ Measuring current

5. In a p-type semiconductor, the major carriers are the?

- A. Electrons
- B. ☒ Holes
- C. Protons
- D. Neutrons

6. The main organic compound formed when bromine water is added to but-1-ene is?

- A. Dibromobutane ☒
- B. Butanol
- C. Butanone
- D. Butanoic acid

7. Platinum electrodes are dipped into  $\text{CuSO}_4$  in a voltmeter. The solution left after electrolysis is

- A. Clear ☒
- B. Blue
- C. Pale blue
- D. Sky blue

8. What volume of water is produced when a mixture of  $150\text{cm}^3$  of hydrogen and  $100\text{cm}^3$  of oxygen is exploded in a eudiometer?

- A.  $50\text{ cm}^3$
- B.  $100\text{ cm}^3$
- C. ☒  $75\text{ cm}^3$
- D.  $150\text{ cm}^3$

9. Oxygen is extracted from water by

- A. Displacement reaction
- B. Reduction reaction
- C. Oxidation reaction
- D. ☒ Decomposition reaction

10. What quantity of current is required to deposit 2.4g of Cu in 750s during electrolysis? ( $\text{Cu}=64$ ,  $F=96500$ )

- A. 0.5A
- B. ☒ 0.32A
- C. 0.28A
- D. 1.2A

11. Which of this requires crystallization mostly?

- A. ☒ Drug making
- B. Cement making
- C. Paint making
- D. Perfume making

12. How many isomer are present in  $\text{C}_6\text{H}_{14}$ ?

- A. 2
- B. 3
- C. 1
- D. ☒ 5

13. The bond in crystalline ammonium chloride are

- A. Covalent and dative bond
- B. Ionic and covalent bond
- C. Ionic, dative and covalent bond
- D. ☒ Ionic, covalent and hydrogen bond

14. Which of the following pollutant is not usually recycled?

- A. Aluminium can
- B. Glass bottle
- C. ☒ Nuclear waste
- D. Paper waste

15. \_\_\_ can be formed as a reddish brown precipitate, malleable metal with good conductor of heat and electricity

- A. K
- B. Mg
- C. ☒ Cu
- D. Al

16. What are the separation techniques involved in obtaining pure Alum from powdered mixture of alum and sand?

- A. Filtration and crystallization ☒
- B. Evaporation
- C. Distillation
- D. Sublimation

17. The compound containing 60% of Oxygen is

- A. ☒ SO<sub>3</sub>
- B. SO<sub>2</sub>
- C. H<sub>2</sub>O
- D. CO<sub>2</sub>

18. Which of the following is used as control rod in nuclear reactor?

- A. ☒ Carbon
- B. Lead
- C. Platinum
- D. Iron

19. Permalloy is composition of Iron and

- A. ☒ Nickel
- B. Copper
- C. Aluminum
- D. Zinc

20. Which of the following elements can react with oxygen to form XO<sub>2</sub>?

- A. Sulphur
- B. Fluorine
- C. Chlorine ☒
- D. Nitrogen

21. The colour obtained when starch-iodide paper is introduced into chlorine jar is

- A. ☒ Blue black
- B. Red
- C. Green
- D. Yellow

22. The only factor that affects the internal energy of an ideal gas is

- A. Pressure
- B. ☒ Temperature
- C. Volume
- D. Mass

23. The compound that decolorizes KMnO<sub>4</sub> and produces effervescence with NaHCO<sub>3</sub> is

- A. ☒ Ethanedioic acid
- B. Ethanoic acid
- C. Methanoic acid
- D. Propanoic acid

24. The reaction between strong acid and strong base produce what type of salt?

- A. ☒ Normal
- B. Acidic
- C. Complex
- D. Basic

25. When acidic oxide dissolves in water it produces

- A. Acid anhydride ☒

- B. Base
- C. Salt
- D. Alkali

26. The shape of CO<sub>2</sub> is

- A. Tetrahedral
- B. ☒ Linear
- C. Trigonal planar
- D. Bent

27. If a jewel is placed at the bottom of a pond 5m deep, calculate the apparent displacement of the jewel when it is viewed directly from above. Refractive index of water =  $\frac{4}{3}$

- A. 3.75 m ☒
- B. 5 m
- C. 4 m
- D. 2.5 m

28. Velocity ratio of two forces inclined at right angles to one another at angle 30° is

- A. 2
- B. ☒ 1.93
- C. 1.5
- D. 1

29. If gas X diffuses twice as fast as gas Y then

- A. Gas X is two times heavier than Y
- B. ☒ Gas Y is 2 times heavier than X
- C. Gas X is four times heavier than Y
- D. Gas Y is four times heavier than X

30. Water glass is made from which compound?

- A. Na<sub>2</sub>SiO<sub>3</sub> ☒
- B. Na<sub>2</sub>CO<sub>3</sub>
- C. K<sub>2</sub>SiO<sub>3</sub>
- D. CaSiO<sub>3</sub>

31. Benzene forms nitrobenzene at 60°C when concentrated HNO<sub>3</sub> reacts with which acid?

- A. ☒ H<sub>2</sub>SO<sub>4</sub>
- B. HCl
- C. H<sub>3</sub>PO<sub>4</sub>
- D. HF

32. What happens when heat is applied to the jointed ends of the wires of a thermocouple?

- A. Voltage is generated ☒
- B. Current stops
- C. Resistance decreases
- D. Wire melts

33. Newton's law of cooling is only valid for

- A. ☒ Small temperature differences
- B. Large temperature differences
- C. Any temperature
- D. Only gases

34. When the vibration of two objects are in tune the best frequency should be equal to

- A. 2 beat per second
- B. 1 beat per second
- C. 0 beat per second ☒
- D. Equal to frequency of vibration of one of the tune

35. Expression for work function is

- A.  $W = hf$
- B. ☒  $W_0 = hf_0$
- C.  $E = E_k$
- D.  $W = E_k$

36. What's the flux density whose flux is 1500 Wb and cross-sectional area of 0.50 m<sup>2</sup>?

- A. 3000 T
- B. 0
- C. 250 T
- D. ☒ 3000 T

37. Combustion of candle produces

- A. CO<sub>2</sub> and H<sub>2</sub>O ☒
- B. CO<sub>2</sub> only
- C. H<sub>2</sub>O only

D. CO only

38. Rust on an iron sheet is

A.  $\text{Fe}_2\text{O}_3$  ✓

B.  $\text{FeO}$

C.  $\text{Fe}_3\text{O}_4$

D.  $\text{Fe}$

39. The reason why coal gas is not used instead of fuel at home is

A. It is expensive

B. It is poisonous ✓

C. Low energy content

D. Difficult to store

40. Function of chlorine in purification of water

A. ✓ Kills bacteria

B. Removes solids

C. Removes color

D. Makes water sweet

41. Force of attraction between molecules of same substance is

A. Intermolecular force ✓

B. Intramolecular force

C. Gravitational force

D. Electrostatic force

42. Calculate the magnetic force if the flux density is 10 T at velocity of  $3 \times 10^7$  m/s at  $60^\circ$  to the magnetic field

A.  $1.5 \times 10^8$  N ✓

B.  $1 \times 10^8$  N

C.  $3 \times 10^7$  N

D.  $5 \times 10^7$  N

43. Binding energy for the nucleon of an element can be calculated using

A. Mass defect and nucleon ✓

B. Atomic number and mass defect

C. Mass defect and mass number

D. Mass number and atomic number

44. Bifocal lens is used to correct

A. Myopia

B. ✓ Presbyopia

C. Hypermetropia

D. Astigmatism

45. Which of these is an application of total internal reflection?

A. ✓ Fiber optics

B. Diverging lens

C. Rainbow

D. Converging lens

46. Given cubic expansivity to be  $3.9 \times 10^{-6}$ , find the area expansivity

A.  $1.3 \times 10^{-6}$  ✓

B.  $3.9 \times 10^{-6}$

C.  $7.8 \times 10^{-6}$

D.  $1.9 \times 10^{-6}$

47. A monochromatic light has how many wavelengths.

A. ✓ One

B. Two

C. Three

D. Many

48. Random change in DNA base is

A. Recombination

B. Hybridization

C. ✓ Mutation

D. Radiation

49. Diffusion is sufficient for transportation in

A. ✓ Earthworm

B. Flatworm

C. Snail

D. Cockroach

50. Endocrine gland called master gland is

A. ✓ Pituitary gland

B. Thyroid gland

C. Adrenal gland

D. Pancreas

1. A wheelbarrow inclined at  $60^\circ$  to the horizontal is pushed with a force of 150 N. What is the horizontal component of the applied force?

- A. 130 N
- B. ☒ 75 N
- C. 60 N
- D. 50 N

$$(150 \times \cos 60^\circ = 150 \times 0.5 = 75 \text{ N})$$

2. Forces that arise as a result of the mobility of electrons include:

- A. Ionic forces
- B. Covalent forces
- C. Hydrogen bonding
- D. ☒ Van der Waals forces

3. Cellulose and starch are classified as

- A. Proteins
- B. Lipids
- C. Nucleic acids
- D. ☒ Carbohydrates

4. Zinc sulphide is used in

- A. ☒ Germicide
- B. Fertilizer
- C. Glass making
- D. Paint

5. The radioisotope used in archaeology is:

- A. Uranium-238
- B. Thorium-232
- C. ☒ Carbon-14
- D. Potassium-40

6. The hydrogenation of benzene to cyclohexane is called

- A. Halogenation
- B. Nitration
- C. Oxidation
- D. ☒ Aromatization

7.  $\text{BaSO}_4$  is

- A. Soluble salt
- B. ☒ Insoluble salt
- C. Acidic salt
- D. Basic salt

8.  $\text{C}_x\text{H}_{12}$ ... what is x in the alkyne of this compound?

- A. 6
- B. ☒ 7
- C. 8
- D. 5

9. Heating of sodalime with  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$  produces

- A. Butane
- B. Propane
- C. ☒ Pentane
- D. Methane

10. The sulphide used in coating electric fluorescent tubes is

- A. Lead sulphide
- B. Copper sulphide
- C. ☒ Zinc sulphide
- D. Iron sulphide

11. Oxidation of a secondary alkanol will give:

- A. ☒ Ketone (e.g., propan-2-ol  $\rightarrow$  propanone)
- B. Alcohol
- C. Aldehyde
- D. Carboxylic acid

12. If the pressure is decreased in the reaction  $\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ , the yield will increase for

- A.  $\text{PCl}_5$
- B. ☒  $\text{PCl}_3$
- C.  $\text{Cl}_2$
- D. None

(Le Chatelier's principle: decreasing pressure favors side with more gas molecules)

13. In the presence of water, anhydrous cobalt(II) chloride turns

- A. Pink to blue
- B. ☒ Blue to pink (hydration changes color)
- C. Red to green
- D. Yellow to blue

14. The Van der Waals' forces are dominant intermolecular forces in

- A. ☒ Ammonium chloride
- B. Water
- C. Alcohol
- D. Hydrogen peroxide

15. The method suitable for separating suspended particles in a liquid is

- A. Filtration
- B. Decantation
- C. ☒ Centrifugation
- D. Evaporation

16. Pressure in a liquid increases with

- A. Depth ☒
- B. Temperature
- C. Volume
- D. Surface area

17. The vertebrae present at the neck region is

- A. Cervical ☒
- B. Thoracic
- C. Lumbar
- D. Sacral

18. The splash zone is found in

- A. Ocean floor
- B. Tide pools
- C. ☒ Intertidal region
- D. River mouth

19. Yellow fever is transmitted by which vector?

- A. ☒ Aedes mosquito

B. Anopheles mosquito

C. Tsetse fly

D. Housefly

20. Wrought iron is basically known as:

- A. ☒ Almost pure iron
- B. Steel
- C. Cast iron
- D. Alloy of iron

21. Which food is digested in the mouth?

- A. Protein
- B. Fat
- C. ☒ Carbohydrates (salivary amylase)
- D. Vitamins

22. The chemical and physical factor of a habitat is known as:

- A. Biotic factor
- B. ☒ Abiotic factor
- C. Ecosystem
- D. Habitat

23. The soil type with the least water retaining capacity is:

- A. Clay
- B. Loamy
- C. ☒ Sandy soil
- D. Peaty soil

24. What is the vegetative part of a plant?

- A. Flower
- B. Seed
- C. ☒ Root, stem, and leaves
- D. Fruit

25. Excretory organ in earthworm is called:

- A. Kidney
- B. Nephridia ☒
- C. Malpighian tubule
- D. Flame cell

26. Breathing through the mouth is called:

- A. Respiration

- B. ☒ Oral respiration
- C. Pulmonary ventilation
- D. Gaseous exchange

27. Water bodies that fluctuate due to movement of salty water into fresh water are:

- A. Rivers
- B. Lakes
- C. ☒ Estuaries
- D. Oceans

28. Cell tissue that leads to primary growth is:

- A. Permanent tissue
- B. Ground tissue
- C. ☒ Apical meristem
- D. Collenchyma

29. The abiotic factor that affect terrestrial habitat is:

- A. Predators
- B. Competition
- C. ☒ Temperature, sunlight, water
- D. Parasites

30. An example of soluble salt is:

- A.  $\text{CaCO}_3$
- B.  $\text{NaCl}$  ☒
- C.  $\text{BaSO}_4$
- D.  $\text{AgCl}$

31. Ammonium gas is dried with:

- A.  $\text{H}_2\text{SO}_4$  ☒
- B.  $\text{NaOH}$
- C. Water
- D. Alcohol

32. \_\_\_\_ is an example of hygroscopic substance:

- A.  $\text{NaCl}$
- B. ☒  $\text{CaCl}_2$
- C. Sugar
- D. Sand

33. Chemical formula for Ester:

- A.  $\text{R-COOH}$
- B.  $\text{R-OH}$
- C. ☒  $\text{R-COOR'}$
- D.  $\text{R-CHO}$

34. Which element can be extracted by the electrolysis of its oxide?

- A. Ca
- B. Na
- C. Mg
- D. ☒ Al

35.  $\text{CH}_3\text{CH}_2(\text{CH}_3)\text{CH}=\text{C}(\text{CH}_3)_2$ . IUPAC name:

- A. 2,3-methylpentene
- B. ☒ 2,3-methylpent-2-ene
- C. 2,3-methylpent-3-ene
- D. 3,4-methylpent-2-ene

36. The number of castes in honey bees:

- A. 5
- B. 4
- C. ☒ 3
- D. 2

Answer: C . (queen, worker, drone)

37. The removal of topsoil by water and wind is called:

- A. ☒ Soil erosion
- B. Soil weathering
- C. Leaching
- D. Gully formation

38. One of the following stores energy in electromagnet:

- A. Transistors
- B. ☒ Inductors
- C. Capacitors
- D. Resistors

39. One of the following is used to measure small amount of resistance:

- A. Voltmeter

- B. Potentiometer
- C. Rheostat
- D. ☒ Wheatstone bridge

40. The frequency of the fourth overtone of a closed pipe is 900 Hz. Find its fundamental frequency:

- A. 100 Hz
- B. ☒ 150 Hz
- C. 200 Hz
- D. 300 Hz

41. An anti-aircraft gun is shot at an aircraft flying with a speed of 335 m/s. Find the distance between them if the gun was shot after 5s that the aircraft passed:

- A. ☒ 1675 m
- B. 335 m
- C. 500 m
- D. 2000 m

Explanation:

(Distance = speed  $\times$  time =  $335 \times 5$ )

42. The evolutionary trend goes as follows:

- A. Bryophyte, thallophyte, pteridophytes and man-fluidatophytes
- B. ☒ Thallophytes, Bryophytes, Pteridophytes and man
- C. Bryophytes, Thallophytes, man, pteridophytes
- D. None

43. The adaptation of brown and green colour of grasshopper is:

- A. Mimicry
- B. Countershading
- C. ☒ Camouflage
- D. Cryptic colouration

44. Which metal is used to make kitchen utensils:

- A. Ca
- B. ☒ Al

- C. Na
- D. K

45. Teams P and Q are involved in a game of football. What is the probability that the game ends in a draw?

- A.  $\frac{2}{3}$
- B.  $\frac{1}{2}$
- C. ☒  $\frac{1}{3}$
- D.  $\frac{1}{4}$

46. Choose the option that has the same consonant sound as Dove:

- A. Gorge
- B. Talked
- C. ☒ Launder
- D. Reached

47. Monoclinic sulphur is obtained by dissolving roll sulphur in:

- A. Methyl-orange
- B. Phenolphthalein
- C. ☒ Methyl benzene
- D. Benzoic acid

48. Electronic configuration of Na

- A. ☒  $1s^2 2s^2 2p^6 3s^1$
- B.  $1s^2 2s^2 2p^5 3s^2$
- C.  $1s^2 2s^2 2p^6 3p^1$
- D.  $1s^2 2s^2 2p^6 3s^2$

49. Molecular shape for carbon(IV)oxide CO<sub>2</sub>

- A. Tetrahedral
- B. Trigonal planar
- C. ☒ Linear
- D. Bent

50. Location of Sudan Savanna

- A. Niger
- B. ☒ Kano
- C. Katsina
- D. Kwara



1. Clumping of red blood cells is known as

- A. ☒ Agglutination
- B. Mutation
- C. Transfusion
- D. Erythroblastosis

2. Countershading is found in:

- A. Crab
- B. Chameleon
- C. ☒ Tilapia
- D. Catfish

3. Which of the following is a continuous variation in man?

- A. Ability to taste PTC
- B. Blood group
- C. ☒ Intelligence
- D. Ability to roll tongue

4. A cross between an albino female and a normal male will lead to which type of offspring?

- A. All genetically normal
- B. ☒ Half phenotypically normal
- C. Half genetically normal
- D. All phenotypically normal

5. Agglutination is likely to occur if O+ is transferred to

- A. ☒ O-
- B. AB
- C. A
- D. B

6. Very large energy band is found in

- A. Conductor
- B. ☒ Insulator
- C. Semiconductor
- D. Extrinsic circuit

7. Kirchoff's junction law is based on the principle of conservation of

- A. Voltage

B. ☒ Current

C. Power

D. Energy

8. Element that provides a mechanism for light to interact with laser lenses is

- A. Glass ☒
- B. Quartz
- C. Aluminium
- D. Silicon

9. Colour of red rose under blue light

- A. Black ☒
- B. Red
- C. Blue
- D. Purple

10. Diaphragm in a camera is similar to which part of the human eye?

- A. ☒ Iris
- B. Retina
- C. Pupil
- D. Lens

11. Vacuum in thermos flask prevents heat loss by

- A. Conduction
- B. ☒ Convection and conduction
- C. Radiation
- D. Reflection

12. Which wave has a parallel direction of propagation?

- A. Transverse
- B. ☒ Longitudinal
- C. Electromagnetic
- D. Surface wave

13. What is the shape of the orbital when the subsidiary quantum number is 2?

- A. s (spherical)
- B. p (dumbbell)
- C. ☒ d (double dumbbell/cloverleaf)
- D. f (complex)

14. In metallurgy, which element is used as a deoxidant?

- A. Carbon
- B. ☒ Aluminium
- C. Silicon
- D. Sulfur

15. The bleaching action of chlorine is due to:

- A. ☒ Oxidation
- B. Reduction
- C. Hydrolysis
- D. Displacement

16. What is used to differentiate between classes of amines?

- A. Tollen's test
- B. ☒ Hinsberg test
- C. Benedict's test
- D. Fehling's test

17. Aluminum is used in the building of vehicles/cars because of:

- A. High density
- B. ☒ Lightweight and corrosion resistance
- C. High melting point
- D. Magnetic properties

18. Phenol-methanal polymer is used to produce:

- A. Nylon
- B. Polyester
- C. ☒ Bakelite
- D. Polyethylene

19. Mathematical expression of Boyle's law:

- A.  $P \propto V$
- B. ☒  $P_1V_1 = P_2V_2$
- C.  $V \propto T$
- D.  $P = nRT$

20. Isomer of  $C_2H_6O$ :

- A. Ethanol ☒
- B. Dimethyl ether

C. Both A and B

D. None

21. What phenomenon is it when a compound loses its water of crystallization?

- A. ☒ Efflorescence
- B. Hygroscopy
- C. Deliquescence
- D. Sublimation

22. Laughing gas is:

- A.  $CO_2$
- B.  $N_2$
- C. ☒  $N_2O$
- D.  $NO_2$

23. When the subsidiary quantum number (l) equals 1, the shape of the orbital is:

- A. s (spherical)
- B. ☒ p (dumbbell)
- C. d (cloverleaf)
- D. f (complex)

24. Substances which remain unchanged in a redox reaction:

- A. Oxidizing agents
- B. Reducing agents
- C. ☒ Catalysts
- D. Electrolytes

25. Example of animal in mangrove swamp:

- A. Squirrel
- B. ☒ Oyster
- C. Zebra
- D. Antelope

26. Universal donor is:

- A. ☒ O
- B. A
- C. B
- D. AB

27. A type of relationship in which one benefits and the other suffers is:

A. Commensalism

B. ☒ Parasitism

C. Predation

D. Mutualism

28. Which of the following is a way of controlling schistosomiasis?

A. ☒ Destroying water snail and water weed

B. Properly washing fruit before eating

C. Clearing bushes around

D. Treating water with chlorine

29. Deleterious effect of overcrowding is:

A. ☒ Disease outbreak

B. Increase predation

C. Rapid dispersion

D. Decrease reproduction

30. Main excretory organ in insects is:

A. Kidney

B. Nephron

C. ☒ Malpighian tubules

D. Nephridia

31. The relationship between man and plasmodium is:

A. Mutualism

B. ☒ Parasitism

C. Commensalism

D. Predation

32. When light is incident on a parallel-sided slab on one side, the shift between the incident and emergent ray is called:

A. Parallel shift

B. Ray shift

C. ☒ Lateral shift

D. Normal shift

33. Millikan experiment was used to determine the:

A. Mass of electron

B. ☒ Charge of electron

C. Mass defect

D. Atomic number

34. If a cell has 75% efficiency and an internal resistance of  $2\Omega$ , find R:

A.  $6\Omega$  ☒

B.  $2\Omega$

C.  $4\Omega$

D.  $8\Omega$

35. The gas used in the demonstration of the fountain experiment is:

A. Hydrogen sulfide

B. Sodium chloride

C. Phosphorus

D. ☒ Hydrogen chloride

36. Formula for ether:

A. ☒  $R-O-R'$

B.  $R-COOH$

C.  $R-CHOH$

D.  $R-CHO$

37. Alkylation of benzene takes place in the presence of:

A.  $H_2SO_4$

B. ☒  $AlCl_3$

C.  $NaOH$

D.  $KMnO_4$

38. "Let the case stand over for the present" means:

A. ☒ Adjourn the case

B. Continue immediately

C. Cancel the case

D. Refer to higher court

39. Nitrogenous wastes in insects first enter the:

A. ☒ Malpighian tubules

B. Nephridia

C. Kidney

D. Nephron

40. Capacitor with  $C = 2F$ ,  $Q = 3C$ . Energy stored = ?

- A. 9 J ☒
- B. 6 J
- C. 12 J
- D. 3 J

41. Register on extinction of languages is used for:

- A. Documentation of languages ☒
- B. Language revival
- C. Translation services
- D. Linguistic research

42. Agglutination occurs when  $O^+$  donates blood to:

- A.  $O^-$
- B. ☒ AB
- C. A
- D. B

43. Which shows that DNA of similar species is more related than distant species:

- A. Fossil records
- B. Microevolution
- C. Comparative anatomy
- D. ☒ Molecular evidence

44. The relationship between man and plasmodium is:

- A. Mutualism
- B. ☒ Parasitism
- C. Commensalism
- D. Predation

45. The 'Notes of Salvation' as used for cheat notes were:

- A. ☒ Euphemistic
- B. Literal
- C. Figurative
- D. None

46. Sclerenchyma tissue consists of:

- A. Parenchyma cells

B. Collenchyma

- C. ☒ Dead thick-walled cells
- D. Meristematic cells

47. Major buffer in blood is:

- A. Phosphate
- B. Proteins
- C. ☒ Bicarbonate ( $HCO_3^-$ )
- D. Amino acids

48. Components of cell sap include:

- A. Water
- B. Sugars
- C. Ions
- D. ☒ All of the above

49. Absorption of food substances in mammalian embryo is via:

- A. Lungs
- B. Skin
- C. ☒ Placenta
- D. Intestine

50. Impulses of smell are received by:

- A. Tongue
- B. Eye
- C. Ear
- D. ☒ Olfactory bulb

1. Infectious diseases are caused by

- A. Hygiene
- B. Microorganisms
- C. Vitamins
- D. Toxins

Answer: B

2. Cooling curves are used to determine

- A. Mass
- B. Boiling point
- C. Melting point
- D. Density

Answer: C

3. The angle of contact between a liquid and container is

- A.  $0^\circ$
- B.  $90^\circ$
- C. Acute or obtuse
- D.  $180^\circ$

Answer: C

4. A wave in which direction of vibration = direction of propagation is

- A. Transverse wave
- B. Stationary wave
- C. Longitudinal wave
- D. Tidal wave

Answer: C

5. For a concave mirror to form a real image, the object must be placed

- A. Between mirror and focus
- B. At the pole
- C. Beyond the focus
- D. At the centre of curvature

Answer: C

6. Ratio of holes to electrons in a pure semiconductor is

- A. 100:1
- B. 10:1
- C. 1:1
- D. 1:10

Answer: C

7. Voltage across two resistors  $7\Omega$  &  $8\Omega$  in series with 150 V supply

- A. 50 V & 60 V
- B. 70 V & 80 V
- C. 75 V & 75 V
- D. 49 V & 101 V

Answer: B  
(Voltage  $\propto$  resistance)

8. Opposite in meaning to: Densely

- A. Thickly
- B. Heavily
- C. Sparsely
- D. Broadly

Answer: C

Motorist (Opposite)

- A. Cyclist
- B. Passenger
- C. Pedestrian
- D. Driver

Answer: C

Broad (Opposite)

- A. Wide
- B. Narrow
- C. Large
- D. Massive

Answer: B

9. Nearest in meaning: Reckless

- A. Careless
- B. Cautious
- C. Intelligent
- D. Safe

Answer: A

Deprecated

- A. Disapproved
- B. Encouraged
- C. Renewed
- D. Respected

Answer: A

Embezzled

- A. Stolen
- B. Increased
- C. Added

D. Deposited

Answer: A

10. Characteristics of loamy soil

A. Poor aeration

B. Too sandy

C. Rich and well-drained

D. Waterlogged

Answer: C

11. Change in gene is

A. Evolution

B. Mutation

C. Variation

D. Fertilization

Answer: B

12. What oxide reacts with water to give an acid

A.  $\text{CO}_2$

B.  $\text{Na}_2\text{O}$

C.  $\text{MgO}$

D.  $\text{CaO}$

Answer: A

13. Halophytes are

A. Desert plants

B. Waterlogged plants

C. Salt-tolerant plants

D. Shade plants

Answer: C

14. A person with blood group B and AB can receive blood from

A. O only

B. A only

C. B and O

D. B, O and A

Answer: D

15. Friction due to air mass can be reduced by

A. Streamlining

B. Propeller

C. Polishing

D. Lubricating

Answer: A

16. Which of these is NOT a semiconductor

A. Silicon

B. Germanium

C. Cadmium

D. Selenium

Answer: C

17. Ability to form 2, 3 or more bonds is called

A. Hybridization

B. Polarisation

C. Allotropy

D. Catenation

Answer: D

18. Reaction between triglyceride and  $\text{NaOH}$  is

A. Polymerization

B. Esterification

C. Saponification

D. Hydrolysis

Answer: C

19. Region with low rainfall & high temperature

A. Sahel savanna

B. Sudan savanna

C. Northern Guinea savanna

D. Southern Guinea savanna

Answer: A

20. Responsible for natural adaptation in organisms

- A. Mutation
- B. Gene flow
- C. Natural selection
- D. Gene pool

Answer: C

21. Northern Guinea Savanna borders

- A. Sahel
- B. Sudan savanna
- C. Mangrove
- D. Rainforest

Answer: B

22. Anemophilous flowers are

- A. Insect-pollinated
- B. Wind-pollinated
- C. Bird-pollinated
- D. Bat-pollinated

Answer: B

23. Lucas test differentiates alkanols based on

- A. Taste
- B. Colour
- C. Rate of substitution
- D. Odour

Answer: C

24. Nucleus is made up of

- A. Electrons only
- B. Protons only
- C. Neutrons only
- D. Protons and neutrons

Answer: D

25.  $\text{H}_2 + \text{Cl}_2$  becomes explosive under

- A. Sunlight
- B. Roman light
- C. Infrared
- D. Ultraviolet

Answer: A

26. Function of swim bladder

- A. Breathing
- B. Feeding
- C. Buoyancy control
- D. Reproduction

Answer: C

27. Benzene chemical properties differ from trienes because of

- A. Linear bonds
- B. Saturation
- C. Delocalized electrons
- D. High reactivity

Answer: C

28. Pericycle of a stem is

- A. Protective layer
- B. Outer cortex
- C. Tissue between endodermis & vascular bundle
- D. Pith region

Answer: C

29. Height of object = 180 m. Velocity halfway (freefall)

- A. 30 m/s
- B. 42 m/s
- C. 20 m/s
- D. 120 m/s

Answer: B

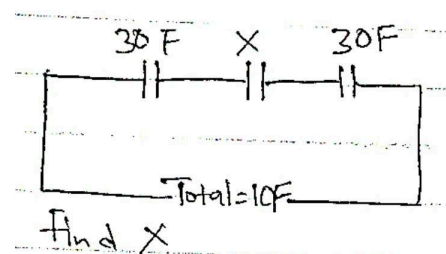
(Using  $v^2 = 2gh$ ,  $h = 90 \text{ m}$ )

30. Increasing temperature of reactant causes

- A. Rate decreases
- B. Rate constant reduces
- C. Rate increases
- D. No effect

Answer: C

31. Find the value of X



32. Hydrocarbon not part of petroleum

- A. Cycloalkanes
- B. Alkanoic acids
- C. Alkanes
- D. Aromatic hydrocarbons

Answer: B

33. Tin reacts with halides to form

- A. Trihalides
- B. Pentahalides
- C. Tetrahalides
- D. Dihalides

Answer: C

34. General formula for ethers

- A.  $\text{RCOOR}'$
- B.  $\text{RCOR}'$
- C.  $\text{ROR}'$
- D.  $\text{RCHO}$

Answer: C

35. Compound giving banana flavour

- A. Alkanoic acids
- B. Alkanals
- C. Alkanoates
- D. Alkenes

Answer: C

36. An emulsion is a colloid of

- A. Solid in solid
- B. Liquid in liquid
- C. Gas in liquid
- D. Solid in gas

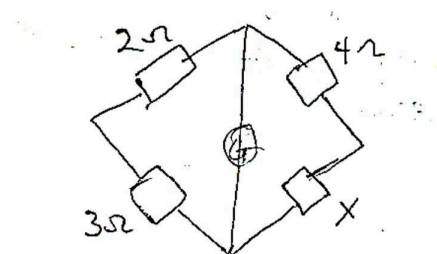
Answer: B

37. Vessel carrying manufactured food

- A. Xylem
- B. Phloem
- C. Cambium
- D. Pith

Answer: B

38. Find the value of X



Handwritten note: Find the value of X

39. Which obeys Pascal's principle

- A. Siphon
- B. Hydraulic press
- C. Syringe
- D. Bicycle pump

Answer: B

40. Unit of inductance is



A. Tesla  
B. Henry  
C. Weber  
D. Ampere  
Answer: B

41. Law that shows direction of induced emf

A. Fleming left hand rule  
B. Lenz law  
C. Faraday's law  
D. Fleming right hand rule  
Answer: B

42. Lenz law is conservation of

A. Matter  
B. Charge  
C. Momentum  
D. Energy  
Answer: D

43. Hormone that controls nerves

A. Cerebellum  
B. Cerebrum  
C. Medulla oblongata  
D. Hypermetropia  
Answer: C

44. When sunlight reflects on the moon before reaching Earth

A. Annular eclipse  
B. Eclipse of the sun  
C. Eclipse of the moon  
D. Transit  
Answer: C

45. Number of ways  $r$  objects can be selected from  $n$  objects

A.  $nPr$

B.  $rPn$   
C.  $nCr$   
D.  $rCn$   
Answer: C

46. Propagation where plant receives food & support from parent

A. Grafting  
B. Layering  
C. Cutting  
D. Budding  
Answer: B

47. Method for separating  $\text{FeCl}_3$  from water + ethanol

A. Evaporation  
B. Filtration  
C. Separating funnel  
D. Distillation  
Answer: C  
(Iron(III) chloride dissolves more in water; ethanol layer separates.)

48. Monomer of polystyrene is

A. Ethene  
B. Phenylethene  
C. Propene  
D. Vinyl chloride  
Answer: B

49. Main difference between benzene & trienes

A. Colour  
B. Aroma  
C. Delocalized electrons  
D. Shape  
Answer: C

50. % of carbon in 5th member of alkanes ( $\text{C}_5\text{H}_{12}$ )

- A. 50%
- B. 66.7%
- C. 83.3%
- D. 91%

Answer: C

1. Which of the following gases is used in the fountain experiment?

- A.  $\text{NH}_3$  only
- B.  $\text{HCl}$  only
- C.  $\text{HCl}$  and  $\text{CO}_2$
- D.  $\text{NH}_3$  and  $\text{HCl}$

Answer: D.  $\text{NH}_3$  and  $\text{HCl}$

2. Which gas forms dense white fumes with ammonia?

- A.  $\text{CO}_2$
- B.  $\text{SO}_2$
- C.  $\text{HCl}$
- D.  $\text{NO}_2$

Answer: C.  $\text{HCl}$

3. Which reaction is common to both saturated and unsaturated hydrocarbons?

- A. Addition reaction
- B. Substitution reaction
- C. Polymerization
- D. Combustion

Answer: D. Combustion

4. A red cloth under red light appears pale because red light \_\_\_\_

- A. Absorbs all colours
- B. Reflects the same colour
- C. Scatters all colours
- D. Refracts the colour

Answer: B. Reflects the same colour

5. After fertilization in plants, the zygote forms the \_\_\_\_

- A. Fruit
- B. Seed
- C. Endosperm
- D. Embryo

Answer: D. Embryo

6. A 5 kg substance melts at constant temperature; this relates to

- A. Latent heat of fusion
- B. Specific latent heat of fusion
- C. Heat capacity
- D. Specific heat capacity

Answer: A. Latent heat of fusion

7. What hormone causes abscission in plants?

- A. Auxin
- B. Gibberellin
- C. Ethylene
- D. Cytokinin

Answer: C. Ethylene

8. Edaphic factors are

- A. Air-related
- B. Soil-related
- C. Water-related
- D. Temperature-related

Answer: B. Soil-related

9. AB blood group is called universal acceptor because it

- A. Has no antigens
- B. Has no antibodies
- C. Has both antibodies
- D. Has only antigen B

Answer: B. Has no antibodies

10. The impurity (phosphine) in the preparation of ethyne is removed by

- A. CaO
- B. Limewater
- C.  $\text{CuSO}_4$
- D. Water

Answer: C.  $\text{CuSO}_4$

11. The acid present in baking powder is

- A. Boric acid
- B. Tartaric acid
- C. Acetic acid
- D. Citric acid

Answer: B. Tartaric acid

12. Water gas is a mixture of

- A.  $\text{CO}_2 + \text{H}_2$
- B.  $\text{CO} + \text{H}_2$
- C.  $\text{CO} + \text{CH}_4$
- D.  $\text{CO}_2 + \text{N}_2$

Answer: B.  $\text{CO} + \text{H}_2$

13. The alkali used in toothpaste is

- A. Calcium hydroxide
- B. Sodium hydroxide
- C. Potassium hydroxide
- D. Magnesium hydroxide

Answer: D. Magnesium hydroxide

14. Copper(II) tetraoxosulphate(VI) ( $\text{CuSO}_4$ ) is used in making

- A. Pigments
- B. Fungicide
- C. Ink
- D. Paint

Answer: B. Fungicide

15. Salivation in hungry dogs in response to a bell is

- A. Nervous action
- B. Autonomic action
- C. Voluntary action
- D. Involuntary action

Answer: B. Autonomic action (conditioned reflex)

16. Exoskeleton made of chitin is an adaptation in

- A. Toad
- B. Snail
- C. Lizard
- D. Insect

Answer: D. Insect

17. Energy of a moving car is

- A. Mechanical
- B. Electrical
- C. Potential
- D. Kinetic

Answer: D. Kinetic

18. The major building block element of organisms is

- A. Nitrogen
- B. Carbon
- C. Water
- D. Sulphur

Answer: B. Carbon

19. Mushrooms consist of a mass of interwoven hyphae called

- A. Stripe
- B. Substratum
- C. Umbrella
- D. Mycelium

Answer: D. Mycelium

20. The acid used in making soft drinks is

- A. Boric
- B. Tartaric
- C. Fatty
- D. Citric

Answer: D. Citric

21. Vertebrate basic body plan similarity shows evolution via

- A. Embryology
- B. Comparative anatomy
- C. Molecular record
- D. Fossil record

Answer: B. Comparative anatomy

22. Vector of malaria is

- A. Tse-tse fly
- B. Female Anopheles mosquito
- C. Housefly
- D. Aedes mosquito

Answer: B. Female Anopheles mosquito

23. The theory of natural selection was postulated by

- A. Lamarck
- B. Darwin
- C. De Vries
- D. Mendel

Answer: B. Darwin

24. The major nutrient required by plants is

- A. Nitrogen
- B. Copper
- C. Lead
- D. Iodine

Answer: A. Nitrogen

1. In the sentence: "Ma, it is not everyone that brings there car to the school," what other term could be used in place of there?

- A. his or her ☒
- B. those
- C. there
- D. that

2. What other business were Stardom school owners involved in?

- A. engineering
- B. manufacturing
- C. real estate
- D. transport ☒

3. The main thing parents admired about Stardom School was?

- A. holding prefect elections
- B. making learning difficult
- C. giving students great exposure ☒
- D. visiting Badagry often

4. What was the relationship between Matthew and Mr. Bepo?

- A. couple
- B. co-workers
- C. co-tenant
- D. family members

5. Choose the word nearest in meaning to pivotal:

"The scientist made a pivotal discovery in cancer research."

- A. crucial ☒
- B. unnecessary
- C. insignificant
- D. minor

6. Choose the word nearest in meaning to esoteric:

"The man enjoys esoteric manipulations."

- A. mystical ☒
- B. basic

- C. dirty
- D. common

7. What was Bepo's answer when Mr. Apeh asked him: "What was the matter, principal?"

- A. He said nothing
- B. He gave details of what was troubling him
- C. He shouted at her
- D. He laughed

8. What does Chief Walimi's story teach us?

- A. many rich people live on loans
- B. many rich people are careless
- C. Stardom management is caring
- D. Stardom School was wonderful

9. What subject does Mrs. Justus Anabel handle at Stardom School?

- A. Mathematics
- B. Chemistry
- C. English
- D. Economics

10. What was the title of the king of Badagry?

- A. Olu of Badagry kingdom
- B. Alaafin of Badagry
- C. Akran of Badagry
- D. Ombadagry of Badagry

11. Choose the word that rhymes with height:

- A. white ☒
- B. weight
- C. eight
- D. hate

12. Choose the option nearest in meaning to vivid:

"They gave a vivid description of the event, making it easy to picture."

- A. clear ☒
- B. brief
- C. confusing
- D. dull

13. Choose the word opposite in meaning to admired:

"The First Lady was admired for her philanthropic gestures."

- A. condemned ☒
- B. cherished
- C. adored
- D. loathed

14. "You again, Mr. Bepo... what brings you out at this time of the day? Is this the students' hostel you are to be minding?"

What figure of speech is used?

- A. rhetorical question ☒
- B. personification
- C. oxymoron
- D. paradox

15. In the sentence below, a word is written in capital letters. Choose the correct interpretation:

"All of US attended the ceremony"

- A. Did all of us criticize the ceremony?
- B. Did all of them attend the ceremony?
- C. Did all of us attend the ceremony? ☒
- D. Did all of us contribute to the ceremony?

16. Choose the word opposite in meaning to jumbled:

"Books are not supposed to be jumbled in the library."

- A. unnumbered
- B. arranged ☒
- C. listed
- D. documented

17. Choose the word that rhymes with town:

- A. own
- B. brown ☒
- C. known
- D. blown

18. In the sentence with emphatic stress: "I will be at the meeting SOON"

Choose the option that relates to it.

- A. Will I be in the meeting soon?
- B. will I be in the meeting soon?
- C. Will I come to the meeting soon? ☒
- D. will I come to the meeting town?

19. Choose the word opposite in meaning to leisurely:

"The old man worked in a leisurely pace."

- A. relaxed
- B. slow
- C. calm
- D. rushed ☒

20. Choose the word opposite in meaning to cruel:

"Someone has to be cruel in order to be..."

- A. kind ☒
- B. respected
- C. honoured
- D. accepted

21. Choose the word with the same vowel sound as edge:

- A. new
- B. bag
- C. jug ☒
- D. blue

22. Who got wrongly accused and was detained in The Lekki Headmaster?

- A. Didi Ogba
- B. Chief Waliem
- C. Fafore
- D. Didi Drogba ☒

23. Choose the word that best completes the gap:

"Nobel Prize for Wole Soyinka was an award for African best..."

- A. literature ☒
- B. writing

- C. culture
- D. game

24. "Protagonist" meaning:

- A. main character ☒
- B. villain
- C. narrator
- D. sidekick

25. "To spill the beans" meaning:

- A. To make a mess
- B. To tell a secret ☒
- C. To cook beans
- D. To deceive

26. "It is high time you PAID" indicates:

- A. past tense
- B. obligation ☒
- C. command
- D. prediction

27. My father together with his friends \_\_\_\_\_ gone.

- A. have
- B. are
- C. has ☒
- D. were

28. The boy as well as his sisters \_\_\_\_\_ intelligent.

- A. are
- B. have
- C. is ☒
- D. were

29. The teacher, with her students, \_\_\_\_\_ gone on excursion.

- A. have
- B. has ☒
- C. is
- D. are

30. The culprit DENIED \_\_\_\_\_ the money.

- A. stealing ☒

- B. to steal
- C. steal
- D. stole

31. Neither the teacher nor the students \_\_\_\_ present.

- A. was ☒
- B. is
- C. are
- D. were

32. I wish I \_\_\_\_ more time to study.

- A. had ☒
- B. have
- C. will have
- D. has

33. The doctor advised that he \_\_\_\_ plenty of water.

- A. drink ☒
- B. drinks
- C. drank
- D. is drinking

34. He has been living here \_\_\_\_ 2001.

- A. since ☒
- B. for
- C. from
- D. at

35. Each of the students \_\_\_\_ given a prize.

- A. was ☒
- B. were
- C. is
- D. have

36. The bride wore a \_\_\_\_ dress.

- A. long
- B. red
- C. beautiful long red ☒
- D. short

37. I am looking forward to \_\_\_\_ you again.

- A. meet

- B. seeing ☒
- C. saw
- D. meets

38. Vowels /i:/ & /ɪ/: choose the word with a different vowel sound:

- A. ship
- B. sheep ☒
- C. bit
- D. sit

39. Which word contains the diphthong /eɪ/?

- A. seat
- B. late ☒
- C. set
- D. bit

40. Which of these has a silent /t/?

- A. Castle ☒
- B. bitter
- C. bottle
- D. tense

41. Vowel sound in bird:

- A. /ɜ:/ ☒
- B. /i:/
- C. /æ/
- D. /ʊ/

42. "The plan fell through" means:

- A. succeeded
- B. postponed
- C. collapsed ☒
- D. canceled

43. "Credulous" means:

- A. skeptical
- B. gullible ☒
- C. confident
- D. cautious

44. "You can't have your cake and eat it" means:

- A. You can have everything

- B. You cannot enjoy two conflicting things ✓  
C. You can share your cake  
D. You must save your cake

45. The power of a lens is measured in?

- A. Joules  
B. Newtons  
C. Diopters ✓  
D. Watts

46. Speed of sound in a solid at 0°C is 100 m/s. Its speed at 25°C is:

- A. 101 m/s ✓  
B. 110 m/s  
C. 125 m/s  
D. 90 m/s

47. A 6V battery delivers current through an external load of 1.25 Ω. What is the power supplied?

- A. 28.8 W ✓  
B. 36 W  
C. 32 W  
D. 25 W

48. According to kinetic theory of gases, pressure exerted by a gas on the wall equals:

- A. force per unit area ✓  
B. volume per unit mass  
C. mass per volume  
D. energy

49. Adding pentavalent impurity atoms to an intrinsic semiconductor produces:

- A. n-type semiconductor ✓  
B. p-type  
C. intrinsic  
D. metallic

50. Which colour in the visible spectrum has the longest wavelength?

- A. Blue  
B. Violet  
C. Yellow

- D. Red ✓

51. Organisms that swarm to avoid overcrowding:

- A. Rats  
B. Termites ✓  
C. Tilapia  
D. Agama lizard

52. Mineral controlling opening/closing of stomata:

- A. Sodium  
B. Phosphorus  
C. Potassium ✓  
D. Molybdenum

53. Dimension of gravitational constant G:

- A.  $M^2LT^{-1}$   
B.  $ML^2T^2$   
C.  $M^{-1}L^3T^{-2}$  ✓  
D.  $M^1L^2T^{-3}$

54. Time for 750W heater to raise 1 kg water from 20°C to 50°C ( $c = 4200 \text{ J/kgK}$ ):

- A. 84 s  
B. 112 s  
C. 168 s ✓  
D. 28 s

55. Which of the following is an epidemic disease?

- A. Malaria  
B. Cholera  
C. Poliomyelitis  
D. Schistosomiasis  
Answer: B (Cholera)

56. Which of the following is important in the evolution of species in a small habitat?

- A. Gene drift  
B. Gene flow



C. Speciation  
D. Biochemical similarities  
Answer: A (Genetic drift)

57. The two-hoofed toes of camels allow easy walking on desert because they have:

A. Long foot surface  
B. Narrow foot surface  
C. Elongated foot surface  
D. Enlarged foot surface  
Answer: D (Enlarged/ broad foot surface)

58. Chemical deodorant is an example of a colloid:

A. Emulsion  
B. Sol  
C. Foam  
D. Aerosol  
Answer: D (Aerosol)

59. Freon pollution in air is released from:

A. Organic decay  
B. Aerosol cans  
C. Photosynthesis  
D. Respiration  
Answer: B (Aerosol cans)

60. Which of these are epidemic diseases in Nigeria?

A. Malaria  
B. Cholera  
C. Schistosomiasis  
D. Typhoid  
Answer: B (Cholera)

61. Structures with similar functions but different origins are:

A. Homologous  
B. Analogous

C. Divergent  
D. Convergent  
Answer: B (Analogous)

62. Pepsin acts in:

A. Alkaline medium  
B. Acidic medium  
C. Both alkaline & acidic  
Answer: B (Acidic medium)

63. Cholera is caused by:

A. Bacteria  
B. Viruses  
C. Fungi  
D. Protozoa  
Answer: A (Bacteria – *Vibrio cholerae*)

64. The most epidemic disease in Nigeria is:

A. Cholera  
B. Malaria  
C. Measles  
D. Typhoid  
Answer: B (Malaria)

65. Absorption of digested food mainly occurs in the:

A. Large intestine  
B. Duodenum  
C. Ileum  
D. Rectum  
Answer: C (Ileum)

66. Which of these did a worker at Stardom School try and fail at?

A. Hawking  
B. Transportation  
C. Farming  
D. Trading  
Answer: A (Hawking)

67. What is the stressed syllable in "Biology"?

- A. Blology
- B. biOlogy
- C. bioLOgy
- D. bioloGY

Answer: C (bioLOgy)

68. How many years remain for Mr. Bepo to retire?

- A. 5
- B. 4
- C. 10
- D. 2

Answer: A (5)

69.. Synonym of affluence

- A. Poverty
- B. Wealth
- C. Laziness
- D. Misfortune

Answer: B (Wealth)

70. Synonym of ruse

- A. Trick
- B. Anger
- C. Peace
- D. Delay

Answer: A

71. Freon pollution is released from:

- A. Organic decay
- B. Aerosol cans
- C. Photosynthesis
- D. Burning wood

Answer: B

72. Empirical formula of 40% C, 6.7% H, 53.3% O:

- A. CHO
- B. CH<sub>2</sub>O
- C. C<sub>2</sub>H<sub>4</sub>O<sub>2</sub>
- D. C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>

Answer: B (CH<sub>2</sub>O)

73. Monomer of polyethylene is:

- A. Ethene
- B. Ethane
- C. Propene
- D. Methane

Answer: A (Ethene)

74. Which metal does NOT liberate H<sub>2</sub> from dilute HCl?

- A. Zinc
- B. Iron
- C. Copper
- D. Magnesium

Answer: C (Copper)

75. Device for measuring atmospheric pressure:

- A. Thermometer
- B. Barometer
- C. Hygrometer
- D. Manometer

Answer: B

76. Compound that decolorizes bromine water:

- A. Ethane
- B. Ethene
- C. Methane
- D. Ethanol

Answer: B

77. Major constituent of air:

- A. Oxygen
- B. Nitrogen
- C. CO<sub>2</sub>
- D. Argon

Answer: B

78. Amphoteric oxide:

- A. Na<sub>2</sub>O
- B. ZnO
- C. CO<sub>2</sub>
- D. SO<sub>2</sub>

Answer: B

79. Major use of H<sub>2</sub>SO<sub>4</sub>:

- A. Fertilizer production
- B. Food seasoning
- C. Water purification
- D. Bleaching

Answer: A

80. Major use of MgSO<sub>4</sub>:

- A. Explosives
- B. Fertilizer
- C. Laxative
- D. Bleaching

Answer: C

81. The distance between two successive crests or troughs is:

- A. Wavelength
- B. Frequency
- C. Amplitude
- D. Period

Answer: A

82. The half-life of a substance is the time taken for its activity to reduce to:

- A. One-quarter

B. One-half

C. Double

D. Zero

Answer: B

83. Specific heat capacity is defined as:

- A. Heat needed to raise 1 kg by 1°C
- B. Heat needed to convert solid to liquid
- C. Heat required to boil water
- D. Heat lost by conduction

Answer: A

84. Moment of force is:

- A. Force × velocity
- B. Mass × distance
- C. Force × perpendicular distance
- D. Work ÷ time

Answer: C

85. Permalloy contains iron and:

- A. Chromium
- B. Nickel
- C. Nitrogen
- D. Oxygen

Answer: B (Nickel)

86. Synonym of indifferent

- A. Caring
- B. Neutral
- C. Angry
- D. Fearful

Answer: B (Neutral)

25. Refractive index of glass = 1.62, water = 1.24. Critical angle =

- A. 50.5°
- B. 30°
- C. 35°
- D. 45°

Answer: A.  $50.5^\circ$

26. When the fulcrum is between the load and effort, the lever is

- A. First class
- B. Second class
- C. Third class
- D. Fourth class

Answer: A. First class

27. Output pin of  $\mu 741$  op-amp is

- A. Pin 2
- B. Pin 3
- C. Pin 4
- D. Pin 6

Answer: D. Pin 6

28. Oxidation number of Mn in MnO and  $\text{MnO}_2$  is

- A. +2 and +7
- B. +3 and +2
- C. +4 and +2
- D. +2 and +4

Answer: D. +2 and +4

29. BCG is used to prevent

- A. Malaria
- B. Tuberculosis
- C. Polio
- D. Cholera

Answer: B. Tuberculosis

30. Accuracy of a vernier caliper is

- A. 0.001 cm
- B. 0.1 cm
- C. 0.05 cm
- D. 0.5 cm

Answer: C. 0.05 cm

31. Ebonite and fur will not have any charge when

- A. Fur transfer charge used to ebonite
- B. Ebonite transfer charges to fur
- C.

Answer: A

32. Ligand in  $[\text{Cu}(\text{NH}_3)_4]^{2+}$  is:

- A.  $\text{NH}_3$
- B. Cu
- C.  $\text{H}_2\text{O}$
- D.  $\text{Cl}^-$

Answer: A

Explanation:  $\text{NH}_3$  donates lone pair.

34. If a linear body changes direction from P to Q, motion is

- A. Rotational
- B. Translational
- C. SHM
- D. Circular

Answer: D. Circular

35. NOT a product of destructive distillation of coal

- A. Coal tar
- B. Phenol
- C. Coke
- D. Coal gas

Answer: B. Phenol

36. Water gas is a mixture of

- A. CO and  $\text{H}_2$
- B.  $\text{CO}_2$  and  $\text{H}_2$
- C. CO and  $\text{N}_2$
- D.  $\text{CO}_2$  and  $\text{H}_2$

Answer: A. CO and  $\text{H}_2$

37. Hydrogenation of ethene produces

- A. Ethyne
- B. Ethanol
- C. Ethane
- D. Propane

Answer: C. Ethane

38. Biome with highest trophic energy

- A. Grassland
- B. Marsh
- C. Forest
- D. Desert

Answer: C. Forest

39. Colour-blind woman × normal man — % of colour-blind sons

- A. 0%
- B. 50%
- C. 100%
- D. 25%

Answer: C. 100%

40. Point of attachment of atlas to skull

- A. Transverse process
- B. Occipital condyles
- C. Pre-zygapophysis
- D. Temporal fusion

Answer: B. Occipital condyles

41. End product of carbohydrate digestion is

- A. Amino acid
- B. Glycerol
- C. Glucose
- D. Sucrose

Answer: C. Glucose

42. NOT a function of blood

- A. Maintaining body temperature

B. Forming clot

C. Distributing bile

D. Transporting excretory materials

Answer: C. Distributing bile

43. Process by which cells divide to produce new cells

- A. Replication
- B. Meiosis
- C. Mitosis
- D. Fertilization

Answer: C. Mitosis

44. Learning to ride a bicycle is controlled by the

- A. Cerebrum
- B. Cerebellum
- C. Medulla
- D. Pons

Answer: B. Cerebellum

45. Effect of catalyst on equilibrium reaction



- A. Increases amount of Z at equilibrium
- B. Increases rate of concentration change
- C. Increases disappearance of X & Y
- D. Increases forward reaction only

Answer: B. Increases rate of reaction (both directions equally)

46. Incomplete oxidation of ethanol yields

- A. Ethanoic acid
- B. Propanone
- C. Diethyl ether
- D. Ethanal

Answer: D. Ethanal

47. Hepatic portal vein is unique because it

- A. Carries deoxygenated blood

- B. Begins and ends in capillaries
- C. Is the largest vein
- D. Carries digested food

Answer: B. Begins and ends in capillaries

48. Orbitals that overlap in methane ( $\text{CH}_4$ )

- A.  $p^2$  and s
- B.  $sp^3$  and s
- C. p and s
- D. s and s

Answer: B.  $sp^3$  and s

49. Number of hydrogen atoms in Aqua Fortis ( $\text{HNO}_3$ )

- A. 2
- B. 1
- C. 3
- D. 4

Answer: B. 1

50.  $\text{H}_2$  can be prepared industrially by

- A. Bosch process
- B. Electrolysis
- C. Action of steam on coke
- D. All of the above

Answer: D

1. In a good thermos flask, heat loss is reduced through the following features EXCEPT?

- A. Lens surface
- B. Insulating cork
- C. Silvered walls
- D. Cork lid

Answer: A. Lens surface

2. Which reaction can undergo thermite process?

- A.  $\text{Zn} + \text{Al}_2\text{O}_3$
- B.  $\text{Mg} + \text{Na}_2\text{O}$
- C.  $\text{Fe} + \text{ZnO}$
- D.  $\text{Mg} + \text{CuO}$

Answer: D.  $\text{Mg} + \text{CuO}$

3. Solubility curve shows variation of solubility with:

- A. Volume
- B. Temperature
- C. Vapour density
- D. Pressure

Answer: B. Temperature

4. The bond between  $\text{H}_2\text{O}$  and  $\text{H}_2$  forming  $\text{H}_3\text{O}^+$  is:

- A. Ionic bond
- B. Hydrogen bond
- C. Coordinate covalent bond
- D. Metallic bond

Answer: C. Coordinate covalent bond

5. In a step-down transformer,  $N_p$  is:

- A.  $= N_s$
- B.  $> N_s$
- C.  $< N_s$
- D.  $\leq N_s$

Answer: B.  $> N_s$

6. During anaerobic respiration in muscles, pyruvic acid is:

- A. Reduced to lactic acid
- B. Oxidized to ethanol
- C. Oxidized to lactic acid
- D. Reduced to water

Answer: A. Reduced to lactic acid

7. Cell wall of prokaryotes is made up of:

- A. Cellulose

- B. Carbohydrate
- C. Protein
- D. Peptidoglycan

Answer: D. Peptidoglycan

8. The man declared his intention from the:
- A. Outset
  - B. Inset
  - C. Onset
  - D. Offset

Answer: A. Outset

9. A property of diamond that qualifies it for jewellery is:
- A. Hardness
  - B. Conductivity
  - C. High refractive index
  - D. Low density

Answer: C. High refractive index

10. The structure that prevents Spirogyra from drying out is:
- A. Chloroplast
  - B. Mucilage sheath
  - C. Pyrenoid
  - D. Nucleus

Answer: B. Mucilage sheath

11. Silvery part of a thermos flask prevents heat loss by:
- A. Conduction
  - B. Convection
  - C. Radiation
  - D. Evaporation

Answer: C. Radiation

12. The number of bones in the middle ear is:
- A. 6
  - B. 3

- C. 5
- D. 8

Answer: B. 3

13. The tissue that insulates the skin is:
- A. Areolar tissue
  - B. Connective tissue
  - C. Adipose tissue
  - D. Epithelial tissue

Answer: C. Adipose tissue

14. A stream flows at 0.75 m/s. A boat heading perpendicular lands at 30°. The boat's velocity is:
- A. 1.00 m/s
  - B. 1.30 m/s
  - C. 1.50 m/s
  - D. 0.75 m/s

Answer: B. 1.30 m/s

15. ---- are good friends.
- A. He and I
  - B. I and him
  - C. I and he
  - D. He and me

Answer: A. He and I

16. African extended family system gives security to ---- members.
- A. His
  - B. Her
  - C. Its
  - D. Their

Answer: D. Their

17. Tayo did not find \_\_\_\_ in the house.
- A. Somebody
  - B. Anyone
  - C. Someone

D. Something

Answer: B. Anyone

18. Point where skull bones meet is the:

- A. Fontanelle
- B. Suture
- C. Joint
- D. Node

Answer: B. Suture

19. Universal donor blood group is:

- A. Type A
- B. Type AB
- C. Type O
- D. Type B

Answer: C. Type O

20. Causative agent of poliomyelitis is

- A. Protozoa
- B. Virus
- C. Bacteria
- D. Worm

Answer: B. Virus

21. Diverging lenses are always:

- A. Convex
- B. Concave
- C. Plano-convex
- D. Cylindrical

Answer: B. Concave

22. Dental formula of man is:

- A.  $i2/2, c2/2, pm2/2, m3/3$
- B.  $i1/1, c2/2, pm2/2, m3/3$
- C.  $i2/2, c1/1, pm3/2, m2/2$
- D.  $i2/2, c1/1, pm2/2, m2/2$
- E.  $i2/2, c1/1, pm2/2, m3/3$

Answer: E.  $i2/2, c1/1, pm2/2, m3/3$

23. Passage of chyme into the duodenum forms:

- A. Digested food
- B. Chymatic product
- C. Chyle
- D. Soluble food
- E. Dissolved food

Answer: C. Chyle

24. Emulsification of fats is caused by

- A. Lipase
- B. Ptyalin
- C. Trypsin
- D. Bile

Answer: D. Bile

25. Every mammal has:

- A. One kidney
- B. Two pairs of kidneys
- C. One pair of kidneys
- D. Four kidneys

Answer: C. One pair of kidneys

26. Faeces is removed in a process called:

- A. Digestion
- B. Egestion
- C. Elimination
- D. Excretion

Answer: B. Egestion

27. Glomerulus is found in the:

- A. Ovary
- B. Testes
- C. Brain
- D. Kidney

Answer: D. Kidney

28. Gas exchange occurs in the:



- A. Trachea
- B. Bronchi
- C. Alveoli
- D. Bronchioles

Answer: C. Alveoli

29. Incisors are used for:

- A. Tearing
- B. Cutting
- C. Grinding
- D. Shearing

Answer: B. Cutting

30. Nematocysts occur in:

- A. Nematodes
- B. Annelids
- C. Platyhelminthes
- D. Cnidarians

Answer: D. Cnidarians

31. Function NOT performed by the skin:

- A. Excretion
- B. Protection
- C. Homeostasis
- D. Reproduction

Answer: D. Reproduction

32. Which alkanol reacts with Lucas reagent?

- A. Primary
- B. Secondary
- C. Tertiary
- D. Quaternary

Answer: C. Tertiary

33. When metals react with air, they form:

- A. Acidic oxides
- B. Basic oxides
- C. Neutral oxides
- D. Amphoteric oxides

Answer: B. Basic oxides

34. Group that reacts vigorously with cold water:

- A. Group 7
- B. Group 4
- C. Group 1
- D. Group 2

Answer: C. Group 1

35. What determines sex in humans?

- A. X chromosome in female
- B. Y chromosome in male
- C. XX in male
- D. YY in female

Answer: B. Y chromosome in male

36. Erwin Schrödinger discovered:

- A. Planetary model
- B. Electron charge
- C. Wave equation for atoms
- D. Proton

Answer: C. Wave equation for atoms

37. Slope of a velocity-time graph is:

- A. Displacement
- B. Speed
- C. Acceleration
- D. Force

Answer: C. Acceleration

38. Pressure in a liquid increases with:

- A. Temperature
- B. Density
- C. Height of liquid column
- D. Volume

Answer: C. Height of liquid column

39. Ore used for electrical wiring is:

- A. Magnetite
- B. Bauxite
- C. Haematite
- D. Lodestone

Answer: B. Bauxite (Aluminium → wires)

40. Tin is extracted from cassiterite by:

- A. Roasting
- B. Reduction
- C. Electrolysis
- D. Carbon reduction

Answer: B. Reduction

41. Shedding of skin is:

- A. Ecdysis
- B. Desquamation
- C. Exfoliation
- D. Dermatitis

Answer: B. Desquamation

42. Molarity of 2%  $\text{H}_2\text{SO}_4$  (MW = 98):

- A. 0.20 M
- B. 0.15 M
- C. 1.0 M
- D. 0.5 M

Answer: A. 0.20 M

43. Decay rate of isotope ( $\lambda = 10^{-8} \text{ s}^{-1}$ ,  $N = 10^{12}$ ):

- A.  $10^4$  decays/s
- B.  $10^6$  decays/s
- C.  $10^5$  decays/s
- D.  $10^7$  decays/s

Answer: A.  $10^4$  decays/s

44. A 50 mA galvanometer ( $2\Omega$ ) converted to 20 V voltmeter requires:

- A.  $398 \Omega$

B.  $878 \Omega$

C.  $908 \Omega$

D.  $100 \Omega$

Answer: C.  $398 \Omega$  series resistor  $\approx 398 \Omega$

45. Major characteristic of the atlas (1st vertebra):

- A. Has odontoid process
- B. Lacks body
- C. Has long spine
- D. Has rib facets

Answer: B. Lacks body

46. Phenol–methanal resin is used for:

- A. Utensil handles
- B. Mattress
- C. Pipes
- D. PVC

Answer: A. Utensil handles

47. Soluble salt is:

- A.  $\text{BaSO}_4$
- B.  $\text{PbCl}_2$
- C.  $\text{NH}_4\text{Cl}$
- D.  $\text{HgCl}_2$

Answer: C.  $\text{NH}_4\text{Cl}$

48. Induced current (Faraday) depends on:

- A. Charge
- B. Mass
- C. Rate of change of flux
- D. Temperature

Answer: C. Rate of change of flux

49. Hot cathode emits electrons by:

- A. Photoelectric effect
- B. Field emission
- C. Thermionic emission
- D. Ionization

Answer: C. Thermionic emission

50. Unit of Volt is:

- A. J/C
- B. W/s
- C. C/J
- D. J·m

Answer: A. J/C

51. n-Heptane is catalysed to give:

- A. Aromatic compounds
- B. Branched isomers
- C. Alcohols
- D. Alkenes

Answer: B. Branched isomers

1. Which of the following is a chemical change?

- A. Melting of ice
- B. Boiling of water
- C. Rusting of iron
- D. Breaking of glass

Answer: C

2. What is the oxidation number of sulfur in  $\text{H}_2\text{SO}_4$ ?

- A. +4
- B. +6
- C. -2
- D. 0

Answer: B

3. Which of the following gases is not a greenhouse gas?

- A.  $\text{CO}_2$
- B.  $\text{CH}_4$
- C.  $\text{N}_2$
- D.  $\text{H}_2\text{O}$  vapour

Answer: C

4. Which particle has the same number of protons and electrons but different number of neutrons?

- A. Isomer
- B. Isotope
- C. Ion
- D. Molecule

Answer: B

5. What is the molar volume of an ideal gas at STP?

- A.  $22.4 \text{ dm}^3$
- B.  $24 \text{ dm}^3$
- C.  $1.0 \text{ dm}^3$
- D.  $2.24 \text{ dm}^3$

Answer: A

6. Which of the following represents a covalent compound?

- A. NaCl
- B. MgO
- C.  $\text{H}_2\text{O}$
- D.  $\text{CaCl}_2$

Answer: C

7. Which of these processes is endothermic?

- A. Freezing of water
- B. Condensation of steam
- C. Dissolution of  $\text{NH}_4\text{Cl}$  in water
- D. Combustion of methane

Answer: C

8. What is the pH of a solution with  $[\text{H}^+] = 1 \times 10^{-4} \text{ mol/dm}^3$ ?

- A. 4
- B. 10
- C. 7
- D. 2

Answer: A

9. What is the IUPAC name of  $\text{CH}_3\text{CH}_2\text{OH}$ ?

- A. Methanol
- B. Ethanol
- C. Propanol

D. Butanol

Answer: B

10. Which of the following is an alkyne?

A.  $C_2H_6$

B.  $C_2H_4$

C.  $C_2H_2$

D.  $CH_4$

Answer: C

11. What is the main ore of aluminum?

A. Hematite

B. Bauxite

C. Galena

D. Magnetite

Answer: B

12. Which of these elements has the highest electronegativity?

A. Fluorine

B. Oxygen

C. Nitrogen

D. Chlorine

Answer: A

13. What is the formula mass of  $Na_2CO_3$ ?

A. 106 g/mol

B. 84 g/mol

C. 40 g/mol

D. 23 g/mol

Answer: A

14. What is the oxidation number of Mn in  $KMnO_4$ ?

A. +2

B. +4

C. +6

D. +7

Answer: D

15. Which of the following is a base?

A.  $H_2SO_4$

B.  $NaOH$

C.  $HCl$

D.  $CO_2$

Answer: B

16. Which of the following shows periodicity?

A. Atomic number

B. Isotopy

C. Mass number

D. Ionization energy

Answer: D

17. Which of the following is not a property of acids?

A. Sour taste

B. Turns blue litmus red

C. Reacts with bases

D. Soapy feel

Answer: D

18. The major constituent of natural gas is:

A. Methane

B. Ethane

C. Butane

D. Propane

Answer: A

19. What is the number of neutrons in  $^{39}_{19}K$ ?

A. 20

B. 19

C. 39

D. 58

Answer: A

20. The process of separating liquids with different boiling points is called:

A. Filtration

B. Crystallization

C. Distillation

D. Sublimation

Answer: C

21. What is the shape of the water molecule?

<https://t.me/iconicimpact>

A. Linear

B. Tetrahedral

- C. Angular (V-shaped)  
D. Trigonal planar

Answer: C

22. Which element forms a basic oxide?

- A.  $\text{CO}_2$   
B.  $\text{SO}_2$   
C.  $\text{Na}_2\text{O}$   
D.  $\text{NO}_2$

Answer: C

23. The number of unpaired electrons in an oxygen atom is:

- A. 0  
B. 1  
C. 2  
D. 4

Answer: C

24. Which of the following has giant covalent structure?

- A.  $\text{NaCl}$   
B.  $\text{H}_2\text{O}$   
C. Diamond  
D.  $\text{CH}_4$

Answer: C

25. Which of the following gases is monoatomic?

- A.  $\text{O}_2$   
B.  $\text{N}_2$   
C. Ne  
D.  $\text{Cl}_2$

Answer: C

26. Which process is used to obtain pure water from salt water?

- A. Filtration  
B. Decantation  
C. Distillation  
D. Crystallization

Answer: C

27. Which of these best describes the solubility of gases in water?

- A. Increases with temperature  
B. Decreases with pressure  
C. Increases with pressure  
D. Independent of pressure

Answer: C

28. Which of the following undergoes addition reaction?

- A. Alkane  
B. Alkene  
C. Alkyne  
D. Aromatic hydrocarbon

Answer: B

29. What is the empirical formula of a compound with 40% C, 6.7% H, and 53.3% O?

- A.  $\text{CHO}$   
B.  $\text{CH}_2\text{O}$   
C.  $\text{C}_2\text{H}_4\text{O}_2$   
D.  $\text{C}_6\text{H}_{12}\text{O}_6$

Answer: B

30. The monomer of polyethylene is:

- A. Ethene  
B. Ethane  
C. Propene  
D. Methane

Answer: A

31. Which of these metals will not liberate hydrogen from dilute  $\text{HCl}$ ?

- A. Zinc  
B. Iron  
C. Copper  
D. Magnesium

Answer: C

32. The device used to measure atmospheric pressure is called:

- A. Thermometer  
B. Barometer

- C. Hygrometer
- D. Manometer

Answer: B

33. The compound that decolorizes bromine water is:

- A. Ethane
- B. Ethene
- C. Methane
- D. Ethanol

Answer: B

34. What is the major constituent of air by volume?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Argon

Answer: B

35. Which of the following is an amphoteric oxide?

- A.  $\text{Na}_2\text{O}$
- B.  $\text{ZnO}$
- C.  $\text{CO}_2$
- D.  $\text{SO}_2$

Answer: B

36. A solution that resists change in pH is a:

- A. Salt
- B. Base
- C. Buffer
- D. Acid

Answer: C

37. Which of these is an extensive property of matter?

- A. Pressure
- B. Temperature
- C. Mass
- D. Density

Answer: C

38. In redox reactions, the oxidizing agent is:

- A. Reduced
- B. Oxidized
- C. A base
- D. Neutral

Answer: A

39. The Tetraoxosulphate(VI) acid is commonly known as:

- A. Nitric acid
- B. Hydrochloric acid
- C. Sulphuric acid
- D. Acetic acid

Answer: C

40. Which of the following is used as a drying agent for acidic gases?

- A.  $\text{NaOH}$
- B.  $\text{CaO}$
- C.  $\text{P}_2\text{O}_5$
- D. Conc.  $\text{H}_2\text{SO}_4$

Answer: D

41. In electrolysis, the electrode at which oxidation occurs is the:

- A. Cathode
- B. Anode
- C. Electrolyte
- D. Salt bridge

Answer: B

42. The product formed when Na reacts with cold water is:

- A.  $\text{NaOH}$  and  $\text{H}_2$
- B.  $\text{Na}_2\text{O}$  and  $\text{H}_2$
- C.  $\text{NaCl}$  and  $\text{H}_2$
- D.  $\text{NaOH}$  and  $\text{O}_2$

Answer: A

43. Which substance contains both ionic and covalent bonds?

- A.  $\text{HCl}$
- B.  $\text{NaCl}$
- C.  $\text{NH}_4\text{Cl}$
- D.  $\text{CH}_4$

Answer: C

44. Which method is suitable for separating a mixture of iron filings and sulfur powder?

- A. Sublimation
- B. Evaporation
- C. Magnetic separation
- D. Filtration

Answer: C

45. What color is litmus in alkaline solution?

- A. Red
- B. Blue
- C. Green
- D. Yellow

Answer: B

46. Which of the following has the highest boiling point?

- A.  $\text{H}_2\text{S}$
- B.  $\text{NH}_3$
- C.  $\text{H}_2\text{O}$
- D.  $\text{CH}_4$

Answer: C

47. What is the electronic configuration of Ca ( $Z=20$ )?

- A. 2,8,8,2
- B. 2,8,10
- C. 2,8,6,4
- D. 2,10,8

Answer: A

48. Which of the following hydrocarbons is saturated?

- A. Ethene
- B. Ethyne
- C. Ethane
- D. Benzene

Answer: C

49. Which halogen is least reactive?

- A. Fluorine
- B. Chlorine

C. Bromine

D. Iodine

Answer: D

50. An isotope of hydrogen with 1 proton and 2 neutrons is:

- A. Protium
- B. Deuterium
- C. Tritium
- D. Hydrogen ion

Answer: C

51. Which of the following is used to prevent rusting?

- A. Galvanization
- B. Distillation
- C. Polymerization
- D. Sublimation

Answer: A

<https://t.me/iconicimpact>

52. The indicator used in strong acid–strong base titration is:

- A. Methyl orange
- B. Phenolphthalein
- C. Litmus
- D. Universal indicator

Answer: B

53. The pH of a neutral solution is:

- A. 14
- B. 7
- C. 0
- D. 1

Answer: B

54. Which of the following is a noble gas?

- A. Nitrogen
- B. Argon
- C. Oxygen
- D. Neon

Answer: B

55. What is the main component of limestone?

- A.  $\text{Na}_2\text{CO}_3$
- B.  $\text{CaCO}_3$
- C.  $\text{CaCl}_2$
- D.  $\text{CaSO}_4$

Answer: B

56. A catalyst increases the rate of a reaction by:

- A. Increasing temperature
- B. Lowering activation energy
- C. Increasing pressure
- D. Changing equilibrium position

Answer: B

57. Which of the following is not an alloy?

- A. Steel
- B. Bronze
- C. Brass
- D. Diamond

Answer: D

58. A salt formed from a strong acid and a strong base is:

- A. Acidic
- B. Basic
- C. Neutral
- D. Amphoteric

Answer: C

59. Which gas turns limewater milky?

- A.  $\text{CO}_2$
- B.  $\text{O}_2$
- C.  $\text{N}_2$
- D.  $\text{CH}_4$

Answer: A

60. An example of a deliquescent substance is:

- A.  $\text{NaCl}$
- B.  $\text{CuSO}_4$
- C.  $\text{NaOH}$
- D.  $\text{CaCO}_3$

Answer: C

61. The movement of particles from a region of higher concentration to a region of lower concentration is called

- A. Diffusion
- B. Filtration
- C. Osmosis
- D. Effusion

Answer: A

62. What is the oxidation state of chlorine in  $\text{NaClO}_3$ ?

- A. +1
- B. -1
- C. +3
- D. +5

Answer: D

63. Which of these is a strong acid?

- A.  $\text{CH}_3\text{COOH}$
- B.  $\text{HNO}_3$
- C.  $\text{H}_2\text{CO}_3$
- D.  $\text{HCN}$

Answer: B

64. Which process involves loss of electrons?

- A. Reduction
- B. Neutralization
- C. Oxidation
- D. Ionization

Answer: C

65. The ability of an atom to attract shared electrons is called

- A. Ionization energy
- B. Electron affinity
- C. Electronegativity
- D. Atomic radius

Answer: C

66. Which of these compounds will give a precipitate with  $\text{AgNO}_3$ ?

- A.  $\text{NaCl}$
- B.  $\text{CH}_4$
- C.  $\text{NH}_3$



D. NaOH

Answer: A

67. The separation technique used to purify ethanol is:

- A. Sublimation
- B. Fractional distillation
- C. Filtration
- D. Decantation

Answer: B

68. Which of the following is not a greenhouse gas?

- A. CH<sub>4</sub>
- B. N<sub>2</sub>O
- C. O<sub>2</sub>
- D. CO<sub>2</sub>

Answer: C

69. The type of bond formed between two nonmetals is

- A. Ionic
- B. Covalent
- C. Metallic
- D. Hydrogen

Answer: B

70. Which substance is added to water to kill germs?

- A. Salt
- B. Lime
- C. Chlorine
- D. Alum

Answer: C

71. The electronic configuration of a chlorine atom (Z=17) is

- A. 2,8,7
- B. 2,7,8
- C. 2,6,9
- D. 2,8,8,1

Answer: A

72. In the periodic table, elements in the same group have

- A. Same number of protons
- B. Similar physical properties
- C. Same number of valence electrons
- D. Same atomic number

Answer: C

73. What is the molecular formula of a compound with empirical formula CH and relative molecular mass of 78?

- A. CH
- B. C<sub>2</sub>H<sub>2</sub>
- C. C<sub>6</sub>H<sub>6</sub>
- D. C<sub>4</sub>H<sub>4</sub>

Answer: C

74. The process of heating ore in the absence of air is called

- A. Roasting
- B. Smelting
- C. Calcination
- D. Alloying

Answer: C

75. What is the color of methyl orange in acid?

- A. Yellow
- B. Orange
- C. Pink
- D. Red

Answer: D

76. Which ion is responsible for the green color of a flame test?

- A. Cu<sup>2+</sup>
- B. Ba<sup>2+</sup>
- C. Fe<sup>3+</sup>
- D. Cr<sup>3+</sup>

Answer: B

77. Which of the following is not a chemical property of an element?

- A. Reactivity

- B. Valency
- C. Boiling point
- D. Ionization energy

Answer: C

78. The formula of sodium trioxocarbonate(IV) is

- A.  $\text{NaHCO}_3$
- B.  $\text{Na}_2\text{CO}_3$
- C.  $\text{NaCl}$
- D.  $\text{NaNO}_3$

Answer: B

79. What is the bond angle in methane ( $\text{CH}_4$ )?

- A.  $109.5^\circ$
- B.  $120^\circ$
- C.  $90^\circ$
- D.  $104.5^\circ$

Answer: A

80. Which of the following is not an organic compound?

- A. Ethanol
- B. Benzene
- C. Glucose
- D.  $\text{NaOH}$

Answer: D

81. Which of these compounds has the lowest boiling point?

- A.  $\text{H}_2\text{O}$
- B.  $\text{NH}_3$
- C.  $\text{CH}_4$
- D.  $\text{HF}$

Answer: C

82. A hydrocarbon with general formula  $\text{C}_n\text{H}_{2n-2}$  is an

- A. Alkane
- B. Alkene
- C. Alkyne
- D. Alcohol

Answer: C

83. The main raw materials for the manufacture of cement are

- A. Limestone and clay
- B. Limestone and coal
- C. Clay and sand
- D. Sand and soda

Answer: A

84. Which of these does not undergo sublimation?

- A. Iodine
- B. Camphor
- C. Sodium chloride
- D. Naphthalene

Answer: C

85. In the electrolysis of acidified water, the gas collected at the cathode is

- A. Oxygen
- B. Hydrogen
- C. Chlorine
- D. Nitrogen

Answer: B

86. Which salt is used in photography?

- A.  $\text{NaCl}$
- B.  $\text{KNO}_3$
- C.  $\text{AgBr}$
- D.  $\text{MgSO}_4$

Answer: C

87. Which compound contains only sigma bonds?

- A. Ethane
- B. Ethene
- C. Ethyne
- D. Benzene

Answer: A

88. The main pollutant in acidic rain is

- A.  $\text{CO}_2$
- B.  $\text{CH}_4$
- C.  $\text{NO}_2$

D.  $\text{H}_2\text{S}$

Answer: C

89. In an endothermic reaction, the enthalpy change is

A. Zero

B. Positive

C. Negative

D. Cannot be determined

Answer: B

90. Which substance is commonly used in the laboratory as a drying agent?

A.  $\text{NaCl}$

B.  $\text{CaCl}_2$

C.  $\text{HCl}$

D.  $\text{CuSO}_4$

Answer: B

91. A reaction in which heat is absorbed from the surroundings is

A. Exothermic

B. Endothermic

C. Neutralization

D. Decomposition

Answer: B

92. Which of the following is an allotrope of carbon?

A. Graphite

B. Carbon dioxide

C. Methane

D. Calcium carbide

Answer: A

93. The major constituent of glass is:

A.  $\text{NaCl}$

B.  $\text{SiO}_2$

C.  $\text{Al}_2\text{O}_3$

D.  $\text{CaO}$

Answer: B

94. A redox reaction involves

A. Loss and gain of protons

B. Loss and gain of electrons

C. Exchange of ions

D. Change in physical state

Answer: B

95. What is the correct IUPAC name of  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ ?

A. Ethanol

B. Methanol

C. Propanol

D. Butanol

Answer: C

96. The hydrocarbon with a triple bond is

A. Ethane

B. Ethene

C. Ethyne

D. Propane

Answer: C

97. Which of the following pollutants is most associated with blood poisoning?

A.  $\text{CO}$

B.  $\text{NO}$

C.  $\text{SO}_2$

D.  $\text{CO}_2$

Answer: A

98. The process used in softening hard water is

A. Filtration

B. Ion exchange

C. Distillation

D. Boiling

Answer: B

99. The main constituent of natural rubber is

A. Polyvinyl chloride

B. Polyethylene

C. Polyisoprene

D. Polystyrene

Answer: C

100. Which of the following is used to detect the end-point in acid-base titration?

- A. Acid
- B. Base
- C. Indicator
- D. Salt

Answer: C

#### QUESTIONS I SAW IN MY JAMB TODAY

1.Monkeys take off with their fore limb and land with their.....

2.The waste product of solvay process is

- A.CaCl<sub>2</sub>
- B.brine
- C.limestone

3.....is used for converting A.C to D.C

- A . Diode
- B. Graphite
- C. Moderator
- D. Capacitors

4.....car is this

- A.who
- B.whose
- C. Whom
- D. Which

5.....is used for photosynthesis in plants

- A .chloroplast
- B. carbon
- C. ribosome
- D. mitochondria

6. Castor oil crops can be grown in which condition?

- A.moderate temperature
- B.mild sunlight
- C. High temperature
- D. Low temperature

7.Anything can be bought here....

- A.can it?
- B.can't it?
- C.can they?

8.Dimension for magnetic field

- A.MT<sup>-2</sup>A
- B.MLT<sup>-2</sup>A<sup>-1</sup>

9.A feasible reaction in which the Gibb's free energy is negative is

- A.spontaneous
- B.exothermic
- C.endothermic
- D.non spontaneous

10.butan-1-ol and pentane are isomers butan-1-ol has boiling points due to the presence of

11. Growth is .....

- A. Irreversible
- B. Reversible

12. Leaves in a fern are called....

- A.fronds
- B.false plants
- C.floral parts

13.To participate in the election the students did one of the following

....

- A. Wrote an essay
- B. The post were placed on the notice board
- C. The sang songs

14. Wool is used in cold regions because

- A. It's traps heat as an insulator
- B. is conducts heat
- C. It is heavy
- D. Weaker materials

Answer: A

15. To prevent resistance the conducting material must be

- A. ☒ Thick
- B. Thin
- C. Hot
- D. Wet

16. The alkaline hydrolysis of tristearin results in the formation of which of the following products?

- A) Primary alcohol
- B) Secondary alcohol
- C) Dihydrochloric acid
- D) Trihydroxy compound

Answer: D

16. Which is an isomer of  $C_4H_{10}$ ?

- A. butane
- B. isobutane
- C. both
- D. neither

Answer: C

Explanation: Two structural forms.

17. Molality uses mass of solvent in:

- A. kg
- B. g
- C. L
- D. mol

Answer: A

Explanation: Definition of molality.

18. Which gas is most soluble in water?

- A.  $NH_3$
- B.  $O_2$
- C.  $N_2$
- D.  $CO_2$

Answer: A

Explanation: Forms H-bonds.

19. Molecular geometry of  $H_2O$  is:

- A. bent
- B. linear
- C. trigonal planar
- D. tetrahedral

Answer: A

Explanation: Two lone pairs.

20. Quantum number  $m_l$  indicates:

- A. orbital orientation
- B. sublevel
- C. spin
- D. principal shell

Answer: A

Explanation: Magnetic quantum number.

21. Beer–Lambert law relates absorbance  $\propto$ :

- A. concentration
- B. volume
- C. mass
- D. pressure

Answer: A

Explanation:  $A = \epsilon lc$ .

22. Rate-determining step is the \_\_\_\_\_ step.

- A. slowest
- B. fastest
- C. last
- D. first

Answer: A

Explanation: Controls overall rate.

23. Mole fraction of solute + solvent =

- A. 1
- B. 0
- C. depends
- D. 2

Answer: A

Explanation: Definition.

24. Covalent bond forms by \_\_\_\_\_ electrons.

- A. sharing
- B. transfer
- C. pooling
- D. losing

Answer: A

Explanation: Definition of covalent bond.

25. Empirical formula from % composition uses:

- A. relative moles
- B. mass
- C. volume
- D. density

Answer: A

Explanation: Simplest whole-number ratio.

26. Saponification produces:

A. soap + glycerol B. ester + acid C. water + ester D. alcohol + ketone

Answer: A

Explanation: Hydrolysis of fat.

27. Rate law from mechanism must match:

A. experimental rate B. theoretical rate C. activation energy D. equilibrium constant

Answer: A

Explanation: Determined empirically.

28. Nernst equation relates potential to:

A. concentration B. temperature C. pressure D. mass

Answer: A

Explanation:  $E = E^\circ - (RT/nF)\ln Q$ .

29. Oxidation involves:

A. loss of electrons B. gain of electrons C. gain of protons D. none

Answer: A

Explanation: OILRIG.

30. Which is amphoteric?

A.  $H_2O$  B.  $NaOH$  C.  $HCl$  D.  $KBr$

Answer: A

Explanation: Can act as acid or base.

31. Bohr model quantizes:

A. electron energy levels B. mass C. size D. spin

Answer: A

Explanation: Discrete orbits.

32. C–C bond length  $\approx$

A. 154 pm B. 100 pm C. 200 pm D. 300 pm

Answer: A

Explanation: Standard single bond length.

33. Hydrogen bonding is strongest between:

A.  $HF$  B.  $HCl$  C.  $HBr$  D.  $HI$

Answer: A

Explanation: High electronegativity of F.

34. Allotrope of carbon that's soft and black is:

A. graphite B. diamond C. fullerene D. graphene

Answer: A

Explanation: Layers slide easily.

35. Claisen condensation involves:

A. esters B. ketones C. aldehydes D. alcohols

Answer: A

Explanation: Two esters form  $\beta$ -keto ester.

36. Which salt is neutral in aqueous solution?

A.  $NaCl$  B.  $NH_4Cl$  C.  $Na_2CO_3$  D.  $AlCl_3$

Answer: A

Explanation: Conjugate acid/base are strong.

37. Complex ion color arises from:

A. d–d transitions B. s–s transitions C. p–p transitions D. f–f transitions

Answer: A

Explanation: Crystal field splitting.

38. Molten  $NaCl$  conducts electricity because:

A. ions are free B. no ions C. electrons free D. neutral molecules

Answer: A

Explanation: Mobile ions carry current.

39. Standard enthalpy of formation of an element =

A. 0 B. -100 kJ C. +100 kJ D. depends

Answer: A

Explanation: By convention.

40. Gibbs free energy  $\Delta G < 0$  means reaction is:

A. spontaneous B. non-spontaneous C. at equilibrium D. impossible

Answer: A

Explanation: Thermodynamically favored.

41. Rate increases with concentration for:

A. elementary reactions B. all reactions C. zero order D. none

Answer: A

Explanation: Depends on molecular collisions.

42. Six-membered aromatic ring is:

A. benzene B. cyclohexane C. cyclobutadiene D. cumulative diene

Answer: A

Explanation:  $6\pi$  electrons aromatic.

43. Ozone decomposes in UV due to:

A. homolytic fission B. heterolytic fission C. ionic bond break D. none

Answer: A

Explanation: Radicals form.

44. Which gas effervesces with acid?

A.  $\text{CO}_2$  B.  $\text{NH}_3$  C.  $\text{O}_2$  D.  $\text{N}_2$

Answer: A

Explanation: Carbonates release  $\text{CO}_2$ .

45. Vapour pressure lowering depends on:

A. mole fraction solute B. mass solute C. volume solute D. density solute

Answer: A

Explanation: Raoult's law.

46. Coordination number in  $[\text{Ni}(\text{NH}_3)_6]^{2+}$  is:

A. 6 B. 4 C. 2 D. 8

Answer: A

Explanation: Six ligands.

47. Which is not chiral?

A. 2-butanol B. lactic acid C. meso-tartaric acid D. alanine

Answer: C

Explanation: Internal plane of symmetry.

48. Molecular formula of glucose is:

A.  $\text{C}_6\text{H}_{12}\text{O}_6$  B.  $\text{C}_5\text{H}_{10}\text{O}_5$  C.  $\text{C}_6\text{H}_6\text{O}_6$  D.  $\text{C}_6\text{H}_{10}\text{O}_5$

Answer: A

Explanation: Common sugar formula.

49. Henry's law relates gas solubility  $\propto$ :

A. partial pressure B. temperature C. volume D. mass

Answer: A

Explanation:  $S = kP$ .

50. Boiling point of water  $\uparrow$  when salt is added because:

A. vapour pressure  $\downarrow$  B. density  $\downarrow$  C. volume  $\downarrow$  D. surface tension  $\downarrow$

Answer: A

Explanation: Colligative effect.

51. Which reagent oxidizes alcohols to ketones?

A. PCC B.  $\text{LiAlH}_4$  C.  $\text{NaBH}_4$  D.  $\text{H}_2/\text{Pd}$

Answer: A

Explanation: PCC mild oxidant.

52. Which is paramagnetic?

A.  $\text{O}_2$  B.  $\text{N}_2$  C.  $\text{CO}_2$  D.  $\text{CH}_4$

Answer: A

Explanation: Two unpaired electrons.

53. Ionization energy generally \_\_\_\_\_ down a group.

A. decreases B. increases C. constant D. varies

Answer: A

Explanation: Atomic radius  $\uparrow$ , IE  $\downarrow$ .

54. Covalent network solid is:

A. diamond B.  $\text{NaCl}$  C. ice D.  $\text{I}_2$

Answer: A

Explanation: Diamond's lattice bonds.

55. Mass defect relates to:

A. nuclear binding energy B. atomic mass unit C. molar mass D. molecular mass

Answer: A

Explanation:  $E = mc^2$ .

56. Which oxide is acidic?

A.  $\text{SO}_3$  B.  $\text{Na}_2\text{O}$  C.  $\text{CaO}$  D.  $\text{ZnO}$

Answer: A

Explanation: Forms  $\text{H}_2\text{SO}_4$  in water.

57. Which gas shows paramagnetism?

A.  $\text{O}_2$  B.  $\text{N}_2$  C.  $\text{CH}_4$  D.  $\text{CO}_2$

Answer: A

Explanation: Unpaired electrons.

58. Which substance is non-electrolyte?

A. sugar B.  $\text{NaCl}$  C.  $\text{HCl}$  D.  $\text{KOH}$

Answer: A

Explanation: Doesn't dissociate.

59. Which molecular orbital is bonding?

A.  $\sigma$  B.  $\sigma^*$  C.  $\pi^*$  D. none

Answer: A

Explanation: Lower energy combination.

60. Ion with electronic config  $[\text{Ar}]3d^{10}$  is:

A.  $\text{Zn}^{2+}$  B.  $\text{Cu}^+$  C. both D. none

Answer: C

Explanation: Both achieve filled d-shell.

1. Site of photosynthesis is:

A. chloroplast B. mitochondrion C. nucleus D. Golgi

Answer: A

Explanation: Contains chlorophyll.

2. Basic unit of life is:

A. cell B. tissue C. organ D. organism

Answer: A

Explanation: Cell theory.

3. Genetic material in viruses is:

A. DNA or RNA B. protein C. lipid D. carbohydrate

Answer: A

Explanation: Viral genomes vary.

4. Mendel's law of segregation refers to separation of:



A. alleles B. chromosomes C. genes D. traits

Answer: A

Explanation: Alleles separate in gametes.

5. Blood vessel with valves is:

A. vein B. artery C. capillary D. arteriole

Answer: A

Explanation: Prevent backflow.

6. Osmosis is movement of water

\_\_\_\_\_ a semipermeable membrane.

A. across B. along C. against D. in

Answer: A

Explanation: Passive diffusion of water.

7. Codon is three nitrogenous bases on:

A. mRNA B. tRNA C. rRNA D. DNA

Answer: A

Explanation: Codes for amino acid.

8. End product of glycolysis is:

A. pyruvate B. acetyl-CoA C. glucose D. lactate

Answer: A

Explanation: Two pyruvate per glucose.

9. Which organ detoxifies drugs?

A. liver B. kidney C. lung D. heart

Answer: A

Explanation: Hepatic detoxification.

10. Xylem transports:

A. water & minerals B. sugars C. amino acids D. hormones

Answer: A

Explanation: Upwards from roots.

11. Which cell junction prevents leaks?

A. tight B. gap C. desmosome D. plasmodesma

Answer: A

Explanation: Seals adjacent cells.

12. Enzyme that unwinds DNA is:

A. helicase B. ligase C. polymerase D. nuclease

Answer: A

Explanation: Separates strands.

13. Blood type universal donor is:

A. O negative B. AB positive C. A positive D. B negative

Answer: A

Explanation: No A/B antigens.

14. Primary site of protein synthesis is:

A. ribosome B. nucleus C. mitochondrion D. ER

Answer: A

Explanation: Ribosomes assemble proteins.

15. Which biogeochemical cycle includes fixation?

A. nitrogen B. carbon C. sulfur D. phosphorus

Answer: A

Explanation:  $N_2 \rightarrow NH_3$  by bacteria.

16. Hormone that raises blood glucose is:

A. glucagon B. insulin C. somatostatin D. adrenaline

Answer: A

Explanation: Stimulates glycogenolysis.

17. Which is a vestigial human organ?  
A. appendix B. liver C. pancreas D. kidney

Answer: A

Explanation: Minimal digestive role.

18. Alleles on same chromosome =  
A. linked B. unlinked C. independent D. dominant

Answer: A

Explanation: Tend to be inherited together.

19. Which organelle contains digestive enzymes?

A. lysosome B. peroxisome C. vacuole D. Golgi

Answer: A

Explanation: Breaks down macromolecules.

20. C4 plants minimize photorespiration via:  
A. spatial separation B. temporal separation C. CAM D. C3 pathway

Answer: A

Explanation: CO<sub>2</sub> pumped to bundle sheath.

21. Blood pH is normally:  
A. 7.4 B. 7.0 C. 8.0 D. 6.8

Answer: A

Explanation: Slightly alkaline.

22. Which phase of mitosis features chromosome alignment?

A. metaphase B. anaphase C. prophase D. telophase

Answer: A

Explanation: Chromosomes on equatorial plate.

23. Which is diploid?

A. somatic cell B. sperm C. egg D. pollen

Answer: A

Explanation: Two sets of chromosomes.

24. Complementary base of cytosine in DNA is:

A. guanine B. adenine C. thymine D. uracil

Answer: A

Explanation: C–G pairing.

25. Which muscle is involuntary?

A. cardiac B. biceps C. triceps D. quadriceps

Answer: A

Explanation: Heart muscle.

26. Trophic level of herbivores is:

A. primary consumer B. producer C. secondary consumer D. decomposer

Answer: A

Explanation: Eat producers.

27. Which gland secretes digestive enzymes into small intestine?

A. pancreas B. liver C. salivary D. thyroid

Answer: A

Explanation: Exocrine function.

28. Red bone marrow produces:

A. blood cells B. bile C. saliva D. hormones

Answer: A

Explanation: Hematopoiesis.

29. Which hormone triggers ovulation?

A. LH B. FSH C. estrogen D. progesterone

Answer: A

Explanation: Luteinizing hormone surge.

30. Which virus causes AIDS?

A. HIV B. HPV C. HCV D. HBV

Answer: A

Explanation: Human immunodeficiency virus.

31. Which disease is caused by protozoan?

A. malaria B. influenza C. measles D. rabies

Answer: A

Explanation: Plasmodium species.

32. Root nodules fix  $N_2$  in:

A. legumes B. grasses C. cereals D. ferns

Answer: A

Explanation: Rhizobia symbiosis.

33. Which is a dihybrid cross ratio?

A. 9:3:3:1 B. 3:1 C. 1:1:1:1 D. 1:2:1

Answer: A

Explanation: Mendel's 2-trait ratio.

34. Which vitamin is water-soluble?

A.  $B_{12}$  B. A C. D D. E

Answer: A

Explanation: B vitamins and C.

35. Peptide bond forms between:

A. amino acids B. nucleotides C. sugars D. fatty acids

Answer: A

Explanation: Between amino and carboxyl groups.

36. Which organelle modifies proteins?

A. Golgi apparatus B. ribosome C. nucleus D. lysosome

Answer: A

Explanation: Packaging and sorting.

37. The light-independent reactions occur in:

A. stroma B. thylakoid C. grana D. cytoplasm

Answer: A

Explanation: Calvin cycle location.

38. Myelin sheath speeds up nerve impulse via:

A. saltatory conduction B. diffusion C. osmosis D. active transport

Answer: A

Explanation: Impulse jumps node-to-node.

39. Which hormone is secreted by adrenal cortex?

A. cortisol B. adrenaline C. insulin D. glucagon

Answer: A

Explanation: Steroid hormone.

40. Which pigment absorbs light for photosynthesis?

A. chlorophyll B. hemoglobin C. melanin D. carotene

Answer: A

Explanation: Main photosynthetic pigment.

41. Which blood cells fight infection?

A. WBC B. RBC C. platelets D. plasma

Answer: A

Explanation: White blood cells.

42. What drives water up xylem?

A. transpiration pull B. root pressure C. capillarity D. all

Answer: D

Explanation: Multiple forces.

43. Which enzyme digests starch?

A. amylase B. protease C. lipase D. nuclease

Answer: A

Explanation: Breaks down polysaccharides.

44. Which nitrogenous base is in RNA but not DNA?

A. uracil B. thymine C. adenine D. guanine

Answer: A

Explanation: Uracil replaces thymine.

45. Which tissue covers body surfaces?

A. epithelial B. connective C. muscle D. nervous

Answer: A

Explanation: Forms barriers and linings.

46. Which structure regulates entry to nucleus?

A. nuclear pores B. ribosomes C. ER D. Golgi

Answer: A

Explanation: Selective transport channels.

47. Which organ system includes pancreas (endocrine)?

A. endocrine B. digestive C. nervous D. respiratory

Answer: A

Explanation: Hormone secretion.

48. Which process produces genetic diversity?

A. crossing-over B. mitosis C. binary fission D. budding

Answer: A

Explanation: Exchange of chromatids.

49. Which enzyme joins Okazaki fragments?

A. DNA ligase B. DNA polymerase C. helicase D. primase

Answer: A

Explanation: Seals nicks in backbone.

50. Primary structure of protein is:

A. amino acid sequence B.  $\alpha$ -helix C.  $\beta$ -pleated sheet D. tertiary fold

Answer: A

Explanation: Linear sequence.

51. Which phylum has poriferans?

A. sponges B. cnidarians C. molluscs D. arthropods

Answer: A

Explanation: Pore-bearing animals.

52. Which gas exchange structure in fish?

A. gills B. lungs C. tracheae D. skin

Answer: A

Explanation: Diffusion across gill filaments.

53. Which blood type has both A and B antigens?

A. AB B. O C. A D. B

Answer: A

Explanation: Universal recipient.

54. Which cell cycle phase is DNA replicated?

A. S B. G<sub>1</sub> C. G<sub>2</sub> D. M

Answer: A

Explanation: Synthesis phase.

55. Which hormone is antagonistic to insulin?

A. glucagon B. somatostatin C. ADH D. oxytocin

Answer: A

Explanation: Raises blood glucose.

56. Which biome is dominated by grasses?

A. savanna B. rainforest C. tundra D. desert

Answer: A

Explanation: Seasonal rainfall.

57. Which organ stores bile?

A. gallbladder B. pancreas C. stomach D. liver

Answer: A

Explanation: Concentrates and stores bile.

58. Which is primary lymphoid organ?

A. bone marrow B. spleen C. lymph node D. tonsil

Answer: A

Explanation: Site of blood cell maturation.

59. Which macromolecule is a polymer of nucleotides?

A. nucleic acid B. protein C. carbohydrate D. lipid

Answer: A

Explanation: DNA and RNA.

60. Which structure produces ATP in prokaryotes?

A. cell membrane B. mitochondrion C. chloroplast D. ER

Answer: A

Explanation: Plasma membrane sites for ETC.

Here are the questions with answers, properly numbered and formatted starting from 1:

1. What is the phenomenon where a wave changes direction as it passes from one medium to another?

A) Reflection  
B) Refraction  
C) Diffraction  
D) Interference

Answer: B) Refraction

2. A 1000 kg car accelerates from rest to 20 m/s in 10 s. What is its average power?

A) 1000 W  
B) 2000 W  
C) 4000 W  
D) 8000 W

Answer: B) 2000 W

3. What is the term for the energy stored in a stretched or compressed spring?

A) Kinetic energy  
B) Potential energy  
C) Thermal energy  
D) Electrical energy

Answer: B) Potential energy

4. A wave has a frequency of 50 Hz and a wavelength of 2 m. What is its speed?

- A) 25 m/s
- B) 50 m/s
- C) 100 m/s
- D) 200 m/s

Answer: C) 100 m/s

5. What is the principle behind the operation of a generator?

- A) Electromagnetic induction
- B) Electromagnetic radiation
- C) Electrostatic induction
- D) Magnetic resonance

Answer: A) Electromagnetic induction

6. A 2  $\mu\text{F}$  capacitor is charged to 100 V. What is the energy stored?

- A) 0.01 J
- B) 0.1 J
- C) 1 J
- D) 10 J

Answer: A) 0.01 J

7. What is the term for the force that opposes motion between two surfaces in contact?

- A) Friction
- B) Gravity
- C) Normal force
- D) Tension

Answer: A) Friction

8. A particle moves with an initial velocity of 5 m/s and accelerates at 2  $\text{m/s}^2$  for 3 s. What is its final velocity?

- A) 10 m/s
- B) 11 m/s
- C) 12 m/s
- D) 15 m/s

Answer: B) 11 m/s

9. What is the term for the ability of a material to return to its original shape after deformation?

- A) Elasticity
- B) Plasticity
- C) Viscosity
- D) Brittleness

Answer: A) Elasticity

10. A 10  $\Omega$  resistor carries a current of 2 A. What is the power dissipated?

- A) 20 W
- B) 40 W
- C) 60 W
- D) 80 W

Answer: B) 40 W

11. What is the term for the minimum energy required to remove an electron from a material?

- A) Work function
- B) Ionization energy
- C) Electron affinity
- D) Binding energy

Answer: A) Work function

12. A block of mass 2 kg is attached to a spring with spring constant 100 N/m. What is its angular frequency?

- A) 5 rad/s
- B) 7 rad/s
- C) 10 rad/s
- D) 14 rad/s

Answer: B) 7 rad/s

13. What is the term for the spreading of waves around obstacles or through small openings?

- A) Reflection
- B) Refraction
- C) Diffraction
- D) Interference

Answer: C) Diffraction

14. A 5 A current flows through a  $10\ \Omega$  resistor. What is the power dissipated?

- A) 100 W
- B) 250 W
- C) 500 W
- D) 1000 W

Answer: B) 250 W

15. An object is in free fall when:

- A) It is thrown upward
- B) It is at rest
- C) The only force acting on it is gravity
- D) It moves down

Answer: C) The only force acting on it is gravity

16. What phenomenon explains why we see different colors in a rainbow?

- A) Reflection
- B) Refraction
- C) Diffraction
- D) Dispersion

Answer: D) Dispersion

17. What is the kinetic energy of an object with a mass of 10 kg moving at a velocity of 5 m/s?

- A) 25 Joules
- B) 50 Joules
- C) 100 Joules
- D) 125 Joules

Answer: D) 125 Joules

18. What is the formula for calculating gravitational potential energy?

- A)  $PE = mgh$
- B)  $PE = \frac{1}{2}mv^2$
- C)  $PE = Fd$
- D)  $PE = mv$

Answer: A)  $PE = mgh$

19. The unit of electric charge is:

- A) Volt
- B) Coulomb
- C) Ampere
- D) Ohm

Answer: B) Coulomb

20. In simple harmonic motion, the acceleration of the object is:

- A) Directly proportional to its displacement
- B) Inversely proportional to its displacement
- C) Constant

Answer: A) Directly proportional to its displacement

21. The phenomenon by which sound waves bend around obstacles is called:

- A) Refraction
- B) Reflection
- C) Diffraction
- D) Interference

Answer: C) Diffraction

22. What is the main source of energy for the Earth?

- A) Electrical Energy
- B) Solar energy
- C) Wind energy
- D) Nuclear energy

Answer: B) Solar energy

23. The process of converting a gas to solid is known as:

- A) Melting
- B) Deposition
- C) Freezing
- D) Condensation

Answer: B) Deposition

24. In a vacuum, all electromagnetic waves travel at:

- A) Different speeds
- B) Light speed
- C) Sound speed
- D) Varying speeds depending on wavelength

Answer: B) Light speed

25. What type of lens is used to correct hyperopia (farsightedness)?

- A) Concave lens
- B) Convex lens
- C) Plane lens
- D) Cylindrical lens

Answer: B) Convex lens

26. What is the IUPAC name of  $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ ?

- A. Benzyl alcohol ☒
- B. Phenol
- C. Ethanol
- D. Cyclohexanol

27. Calculate the number of moles in 50 g of NaCl ( $M = 58.5 \text{ g/mol}$ ).

- A. 0.85 mol
- B. 0.50 mol
- C. 0.68 mol ☒
- D. 1.20 mol

28. Oxidation number of sulfur in  $\text{H}_2\text{SO}_4$  is:

- A. +4
- B. +6 ☒
- C. -2
- D. +2

29. Molarity of 5 g NaOH in  $200 \text{ cm}^3$  solution ( $\text{NaOH} = 40 \text{ g/mol}$ ):

- A. 0.25 M
- B. 0.50 M
- C. 0.75 M
- D. 0.625 M ☒

30. Main product when an alkene undergoes hydration with conc.  $\text{H}_2\text{SO}_4$ :

- A. Alkane
- B. Alcohol ☒
- C. Aldehyde
- D. Ether

31. Molar volume of a gas at STP:

- A.  $11.2 \text{ dm}^3$
- B.  $22.4 \text{ dm}^3$  ☒
- C.  $24.0 \text{ dm}^3$
- D.  $44.8 \text{ dm}^3$

32. Element with highest electronegativity:

- A. Oxygen
- B. Nitrogen
- C. Chlorine
- D. Fluorine ☒

33. Instrument used to measure diameter of a thin wire:

- A. Metre rule
- B. Vernier caliper
- C. Screw gauge ☒



D. Tape rule

34. SI unit of pressure is:

- A. Watt
- B. Pascal ✓
- C. Joule
- D. Newton

35. Which of the following is a vector quantity?

- A. Speed
- B. Distance
- C. Mass
- D. Force ✓

36. Resultant of 5 N and 12 N at right angles

- A. 13 N ✓
- B. 17 N
- C. 7 N
- D. 12.5 N

37. A car accelerates from rest to 20 m/s in 5 s. Acceleration:

- A. 2 m/s<sup>2</sup>
- B. 4 m/s<sup>2</sup> ✓
- C. 5 m/s<sup>2</sup>
- D. 20 m/s<sup>2</sup>

38. Area under a velocity-time graph represents:

- A. Acceleration
- B. Force
- C. Distance/Displacement ✓
- D. Speed

39. Work is defined as:

- A. Force per unit distance

B. Force × displacement in the direction of force ✓

C. Power × time

D. Energy per time

40. A machine with 80% efficiency receives 200 J input. Output = ?

- A. 160 J ✓
- B. 180 J
- C. 40 J
- D. 100 J

41. A good conductor of heat is:

- A. Glass
- B. Plastic
- C. Copper ✓
- D. Wood

42. Heat required to raise 1 kg of a substance by 1°C is:

- A. Latent heat
- B. Specific latent heat
- C. Heat capacity
- D. Specific heat capacity ✓

43. Bending of light when passing different media is:

- A. Reflection
- B. Refraction ✓
- C. Diffraction
- D. Interference

44. A real image is formed by a convex lens when object is placed:

- A. Between lens and focal point
- B. At focal point
- C. Beyond 2F ✓
- D. Between F and 2F

45. Example of a third class lever:

- A. Claw hammer
- B. Scissors
- C. Wheel barrow
- D. Sugar tongs ✓

46. Floatation EXCEPT:

- A. Upthrust equals weight
- B. Floating object displaces its weight
- C. Objects float if density is less
- D. Object floats because weight of fluid is more than object ✓

47. Water heater: time taken by 750 W heater to raise 1 kg water from 20°C to 50°C ( $c = 4200 \text{ J/kg}^\circ\text{C}$ ):

- A. 84 s
- B. 112 s
- C. 168 s ✓
- D. 28 s

48. Test for  $\text{Cu}^{2+}$  using NaOH produces:

- A.  $2\text{Cu}^{2+} + \text{NaOH}$
- B.  $\text{Cu}(\text{OH})_2 + 2\text{Na}^+$  (blue ppt.) ✓
- C.  $\text{NaOH} + \text{Cu}$
- D.  $\text{Cu}(\text{OH})_2 + 2\text{NaH}$

49. Dimension of G (Gravitational constant):

- A.  $\text{M}^2\text{LT}^{-1}$
- B.  $\text{ML}^2\text{T}^2$
- C.  $\text{M}^1\text{L}^2\text{T}^{-3}$
- D.  $\text{M}^{-1}\text{L}^3\text{T}^{-2}$  ✓

50. Stress pattern of "INEQUALITY":

- A. IN-e-qua-li-ty
- B. in-E-qua-li-ty
- C. in-e-QUA-li-ty ✓
- D. in-e-qua-LI-ty

E. in-e-qua-li-TY

51. Lens used to correct hyperopia:

- A. Concave
- B. Convex ✓
- C. Plane
- D. Cylindrical

52. Unit of Force:

- A. Joule
- B. Newton ✓
- C. Pascal
- D. Watt

53. Pressure is defined as:

- A. Force per unit area ✓
- B. Mass per volume
- C. Energy per time
- D. Work per distance

54. Phenomenon where particle crosses energy barrier:

- A. Tunneling ✓
- B. Reflection
- C. Refraction
- D. Diffraction

55. KE of 10 kg moving at 5 m/s:

- A. 125 J ✓
- B. 50 J
- C. 100 J
- D. 75 J

56. Relationship between rabbit & bacteria:

- A. Mutualism ✓
- B. Commensalism
- C. Parasitism
- D. Amensalism

57. Sol and gel are examples of:

- A. Suspension
- B. Colloid ☒
- C. Aerosol
- D. Emulsion

58. Gases used in fountain experiment:

- A.  $\text{CO}_2$  &  $\text{NH}_3$
- B.  $\text{NH}_3$  &  $\text{HCl}$  ☒
- C.  $\text{H}_2$  &  $\text{O}_2$
- D.  $\text{H}_2$  &  $\text{Cl}_2$

59. Where does glycolysis occur?

- A. Nucleus
- B. Mitochondria
- C. Cytosol (cytoplasm) ☒
- D. Ribosome

60. Powerhouse of the cell:

- A. Chloroplast
- B. Ribosome
- C. Mitochondria ☒
- D. Nucleus

61. Excretory organ of an earthworm:

- A. Kidney
- B. Nephridium ☒
- C. Malpighian tubule
- D. Flame cell

62. Oxidation number of oxygen in  $\text{H}_2\text{O}_2$ :

- A. -2
- B. -1 ☒
- C. 0
- D. +1

63. Synonym of "Esoteric":

- A. Common
- B. Mysterious
- C. Secret/Obscure/Profound ☒
- D. Simple

64. Synonym of "Austere":

- A. Harsh/Strict ☒
- B. Exciting
- C. Colourful
- D. Gentle

65. Synonym of "Onerous":

- A. Easy
- B. Joyful
- C. Difficult/Burdensome ☒
- D. Slow

66. Synonym of "Incentive":

- A. Reward/Motivation ☒
- B. Punishment
- C. Barrier
- D. Attitude

67. Synonym of "Euphoria":

- A. Sadness
- B. Confusion
- C. Intense happiness/joy ☒
- D. Anger

1. In the sentence: "Ma, it is not everyone that brings there car to the school," what other term could be used in place of there?

- A. his or her ☒
- B. those
- C. there
- D. that

2. What other business were Stardom school owners involved in?

- A. engineering
- B. manufacturing
- C. real estate
- D. transport ☒

3. The main thing parents admired about Stardom School was?

- A. holding prefect elections
- B. making learning difficult
- C. giving students great exposure ☒
- D. visiting Badagry often

4. What was the relationship between Matthew and Mr. Bepo?

- A. couple
- B. co-workers
- C. co-tenant
- D. family members

5. Choose the word nearest in meaning to pivotal:

"The scientist made a pivotal discovery in cancer research."

- A. crucial ☒
- B. unnecessary
- C. insignificant
- D. minor

6. Choose the word nearest in meaning to esoteric:

"The man enjoys esoteric manipulations."

- A. mystical ☒
- B. basic
- C. dirty
- D. common

7. What was Bepo's answer when Mr. Apeh asked him: "What was the matter, principal?"

- A. He said nothing
- B. He gave details of what was troubling him
- C. He shouted at her

D. He laughed

8. What does Chief Walimi's story teach us?

- A. many rich people live on loans
- B. many rich people are careless
- C. Stardom management is caring
- D. Stardom School was wonderful

9. What subject does Mrs. Justus Anabel handle at Stardom School?

- A. Mathematics
- B. Chemistry
- C. English
- D. Economics

10. What was the title of the king of Badagry?

- A. Olu of Badagry kingdom
- B. Alaafin of Badagry
- C. Akran of Badagry
- D. Ombadagry of Badagry

11. Choose the word that rhymes with height:

- A. white ☒
- B. weight
- C. eight
- D. hate

12. Choose the option nearest in meaning to vivid:

"They gave a vivid description of the event, making it easy to picture."

- A. clear ☒
- B. brief
- C. confusing
- D. dull

13. Choose the word opposite in meaning to admired:

"The First Lady was admired for her philanthropic gestures."

- A. condemned ☒
- B. cherished
- C. adored

D. loathed

14. "You again, Mr. Bepo... what brings you out at this time of the day? Is this the students' hostel you are to be minding?"

What figure of speech is used?

A. rhetorical question ☒

B. personification

C. oxymoron

D. paradox

15. In the sentence below, a word is written in capital letters. Choose the correct interpretation:

"All of US attended the ceremony"

A. Did all of us criticize the ceremony?

B. Did all of them attend the ceremony?

C. Did all of us attend the ceremony? ☒

D. Did all of us contribute to the ceremony?

16. Choose the word opposite in meaning to jumbled:

"Books are not supposed to be jumbled in the library."

A. unnumbered

B. arranged ☒

C. listed

D. documented

17. Choose the word that rhymes with town:

A. own

B. brown ☒

C. known

D. blown

18. In the sentence with emphatic stress: "I will be at the meeting SOON"

Choose the option that relates to it.

A. Will I be in the meeting soon?

B. will I be in the meeting soon?

C. Will I come to the meeting soon? ☒

D. will I come to the meeting town?

19. Choose the word opposite in meaning to leisurely:

"The old man worked in a leisurely pace."

A. relaxed

B. slow

C. calm

D. rushed ☒

20. Choose the word opposite in meaning to cruel:

"Someone has to be cruel in order to be..."

A. kind ☒

B. respected

C. honoured

D. accepted

21. Choose the word with the same vowel sound as edge:

A. new

B. bag

C. jug ☒

D. blue

22. Who got wrongly accused and was detained in The Lekki Headmaster?

A. Didi Ogba

B. Chief Waliem

C. Fafore

D. Didi Drogba ☒

23. Choose the word that best completes the gap:

"Nobel Prize for Wole Soyinka was an award for African best..."

A. literature ☒

B. writing

C. culture

D. game

24. "Protagonist" meaning:

A. main character ☒

B. villain

C. narrator

D. sidekick

25. "To spill the beans" meaning:

- A. To make a mess
- B. To tell a secret ☒
- C. To cook beans
- D. To deceive

26. "It is high time you PAID" indicates:

- A. past tense
- B. obligation ☒
- C. command
- D. prediction

27. My father together with his friends \_\_\_\_\_ gone.

- A. have
- B. are
- C. has ☒
- D. were

28. The boy as well as his sisters \_\_\_\_\_ intelligent.

- A. are
- B. have
- C. is ☒
- D. were

29. The teacher, with her students, \_\_\_\_\_ gone on excursion.

- A. have
- B. has ☒
- C. is
- D. are

30. The culprit DENIED \_\_\_\_\_ the money.

- A. stealing ☒
- B. to steal
- C. steal
- D. stole

31. Neither the teacher nor the students \_\_\_\_\_ present.

- A. was ☒
- B. is

- C. are
- D. were

32. I wish I \_\_\_\_\_ more time to study.

- A. had ☒
- B. have
- C. will have
- D. has

33. The doctor advised that he \_\_\_\_\_ plenty of water.

- A. drink ☒
- B. drinks
- C. drank
- D. is drinking

34. He has been living here \_\_\_\_\_ 2001.

- A. since ☒
- B. for
- C. from
- D. at

35. Each of the students \_\_\_\_\_ given a prize.

- A. was ☒
- B. were
- C. is
- D. have

36. The bride wore a \_\_\_\_\_ dress.

- A. long
- B. red
- C. beautiful long red ☒
- D. short

37. I am looking forward to \_\_\_\_\_ you again.

- A. meet
- B. seeing ☒
- C. saw
- D. meets

38. Vowels /i:/ & /ɪ/: choose the word with a different vowel sound:

- A. ship
- B. sheep ☒

- C. bit
- D. sit

39. Which word contains the diphthong /eɪ/?

- A. seat
- B. late ✓
- C. set
- D. bit

40. Which of these has a silent /t/?

- A. Castle ✓
- B. bitter
- C. bottle
- D. tense

41. Vowel sound in bird:

- A. /ɜ:/ ✓
- B. /i:/
- C. /æ/
- D. /ʊ/

42. "The plan fell through" means:

- A. succeeded
- B. postponed
- C. collapsed ✓
- D. canceled

43. "Credulous" means:

- A. skeptical
- B. gullible ✓
- C. confident
- D. cautious

44. "You can't have your cake and eat it" means:

- A. You can have everything
- B. You cannot enjoy two conflicting things ✓
- C. You can share your cake
- D. You must save your cake

45. The power of a lens is measured in?

- A. Joules
- B. Newtons
- C. Diopters ✓

D. Watts

46. Speed of sound in a solid at 0°C is 100 m/s. Its speed at 25°C is:

- A. 101 m/s ✓
- B. 110 m/s
- C. 125 m/s
- D. 90 m/s

47. A 6V battery delivers current through an external load of 1.25 Ω. What is the power supplied?

- A. 28.8 W ✓
- B. 36 W
- C. 32 W
- D. 25 W

48. According to kinetic theory of gases, pressure exerted by a gas on the wall equals:

- A. force per unit area ✓
- B. volume per unit mass
- C. mass per volume
- D. energy

49. Adding pentavalent impurity atoms to an intrinsic semiconductor produces:

- A. n-type semiconductor ✓
- B. p-type
- C. intrinsic
- D. metallic

50. Which colour in the visible spectrum has the longest wavelength?

- A. Blue
- B. Violet
- C. Yellow
- D. Red ✓

51. Organisms that swarm to avoid overcrowding:

- A. Rats
- B. Termites ✓
- C. Tilapia
- D. Agama lizard

52. Mineral controlling opening/closing of stomata:

- A. Sodium
- B. Phosphorus
- C. Potassium ✓
- D. Molybdenum

53. Dimension of gravitational constant G:

- A.  $M^2LT^{-1}$
- B.  $ML^2T^2$
- C.  $M^{-1}L^3T^{-2}$  ✓
- D.  $M^1L^2T^{-3}$

54. Time for 750W heater to raise 1 kg water from 20°C to 50°C ( $c = 4200 \text{ J/kgK}$ ):

- A. 84 s
- B. 112 s
- C. 168 s ✓
- D. 28 s

55. Which of the following is an epidemic disease?

- A. Malaria
  - B. Cholera
  - C. Poliomyelitis
  - D. Schistosomiasis
- Answer: B (Cholera)

56. Which of the following is important in the evolution of species in a small habitat?

- A. Gene drift
  - B. Gene flow
  - C. Speciation
  - D. Biochemical similarities
- Answer: A (Genetic drift)

57. The two-hoofed toes of camels allow easy walking on desert because they have:

- A. Long foot surface

- B. Narrow foot surface
- C. Elongated foot surface
- D. Enlarged foot surface

Answer: D (Enlarged/ broad foot surface)

58. Chemical deodorant is an example of a colloid:

- A. Emulsion
  - B. Sol
  - C. Foam
  - D. Aerosol
- Answer: D (Aerosol)

59. Freon pollution in air is released from:

- A. Organic decay
  - B. Aerosol cans
  - C. Photosynthesis
  - D. Respiration
- Answer: B (Aerosol cans)

60. Which of these are epidemic diseases in Nigeria?

- A. Malaria
  - B. Cholera
  - C. Schistosomiasis
  - D. Typhoid
- Answer: B (Cholera)

61. Structures with similar functions but different origins are:

- A. Homologous
  - B. Analogous
  - C. Divergent
  - D. Convergent
- Answer: B (Analogous)

62. Pepsin acts in:

- A. Alkaline medium
- B. Acidic medium



C. Both alkaline & acidic  
Answer: B (Acidic medium)

63. Cholera is caused by:

- A. Bacteria
- B. Viruses
- C. Fungi
- D. Protozoa

Answer: A (Bacteria – *Vibrio cholerae*)

64. The most epidemic disease in Nigeria is:

- A. Cholera
- B. Malaria
- C. Measles
- D. Typhoid

Answer: B (Malaria)

65. Absorption of digested food mainly occurs in the:

- A. Large intestine
- B. Duodenum
- C. Ileum
- D. Rectum

Answer: C (Ileum)

66. Which of these did a worker at Stardom School try and fail at?

- A. Hawking
- B. Transportation
- C. Farming
- D. Trading

Answer: A (Hawking)

67. What is the stressed syllable in "Biology"?

- A. Blology
- B. biOlogy
- C. bioLOgy
- D. bioloGY

Answer: C (bioLOgy)

68. How many years remain for Mr. Bepo to retire?

- A. 5
- B. 4
- C. 10
- D. 2

Answer: A (5)

69.. Synonym of affluence

- A. Poverty
- B. Wealth
- C. Laziness
- D. Misfortune

Answer: B (Wealth)

70. Synonym of ruse

- A. Trick
- B. Anger
- C. Peace
- D. Delay

Answer: A

71. Freon pollution is released from:

- A. Organic decay
- B. Aerosol cans
- C. Photosynthesis
- D. Burning wood

Answer: B

72. Empirical formula of 40% C, 6.7% H, 53.3% O:

- A. CHO
- B. CH<sub>2</sub>O
- C. C<sub>2</sub>H<sub>4</sub>O<sub>2</sub>
- D. C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>

Answer: B (CH<sub>2</sub>O)

73. Monomer of polyethylene is:

- A. Ethene
- B. Ethane
- C. Propene
- D. Methane

Answer: A (Ethene)

74. Which metal does NOT liberate  $H_2$  from dilute HCl?

- A. Zinc
- B. Iron
- C. Copper
- D. Magnesium

Answer: C (Copper)

75. Device for measuring atmospheric pressure:

- A. Thermometer
- B. Barometer
- C. Hygrometer
- D. Manometer

Answer: B

76. Compound that decolorizes bromine water:

- A. Ethane
- B. Ethene
- C. Methane
- D. Ethanol

Answer: B

77. Major constituent of air:

- A. Oxygen
- B. Nitrogen
- C.  $CO_2$
- D. Argon

Answer: B

78. Amphoteric oxide:

- A.  $Na_2O$
- B.  $ZnO$
- C.  $CO_2$
- D.  $SO_2$

Answer: B

79. Major use of  $H_2SO_4$ :

- A. Fertilizer production
- B. Food seasoning
- C. Water purification
- D. Bleaching

Answer: A

80. Major use of  $MgSO_4$ :

- A. Explosives
- B. Fertilizer
- C. Laxative
- D. Bleaching

Answer: C

81. The distance between two successive crests or troughs is:

- A. Wavelength
- B. Frequency
- C. Amplitude
- D. Period

Answer: A

82. The half-life of a substance is the time taken for its activity to reduce to:

- A. One-quarter
- B. One-half
- C. Double
- D. Zero

Answer: B

83. Specific heat capacity is defined as:

- A. Heat needed to raise 1 kg by  $1^\circ C$
- B. Heat needed to convert solid to liquid

- C. Heat required to boil water
- D. Heat lost by conduction

Answer: A

84. Moment of force is:

- A. Force  $\times$  velocity
- B. Mass  $\times$  distance
- C. Force  $\times$  perpendicular distance
- D. Work  $\div$  time

Answer: C

85. Permalloy contains iron and:

- A. Chromium
- B. Nickel
- C. Nitrogen
- D. Oxygen

Answer: B (Nickel)

86. Synonym of indifferent

- A. Caring
- B. Neutral
- C. Angry
- D. Fearful

Answer: B (Neutral)